

Project Data Page:

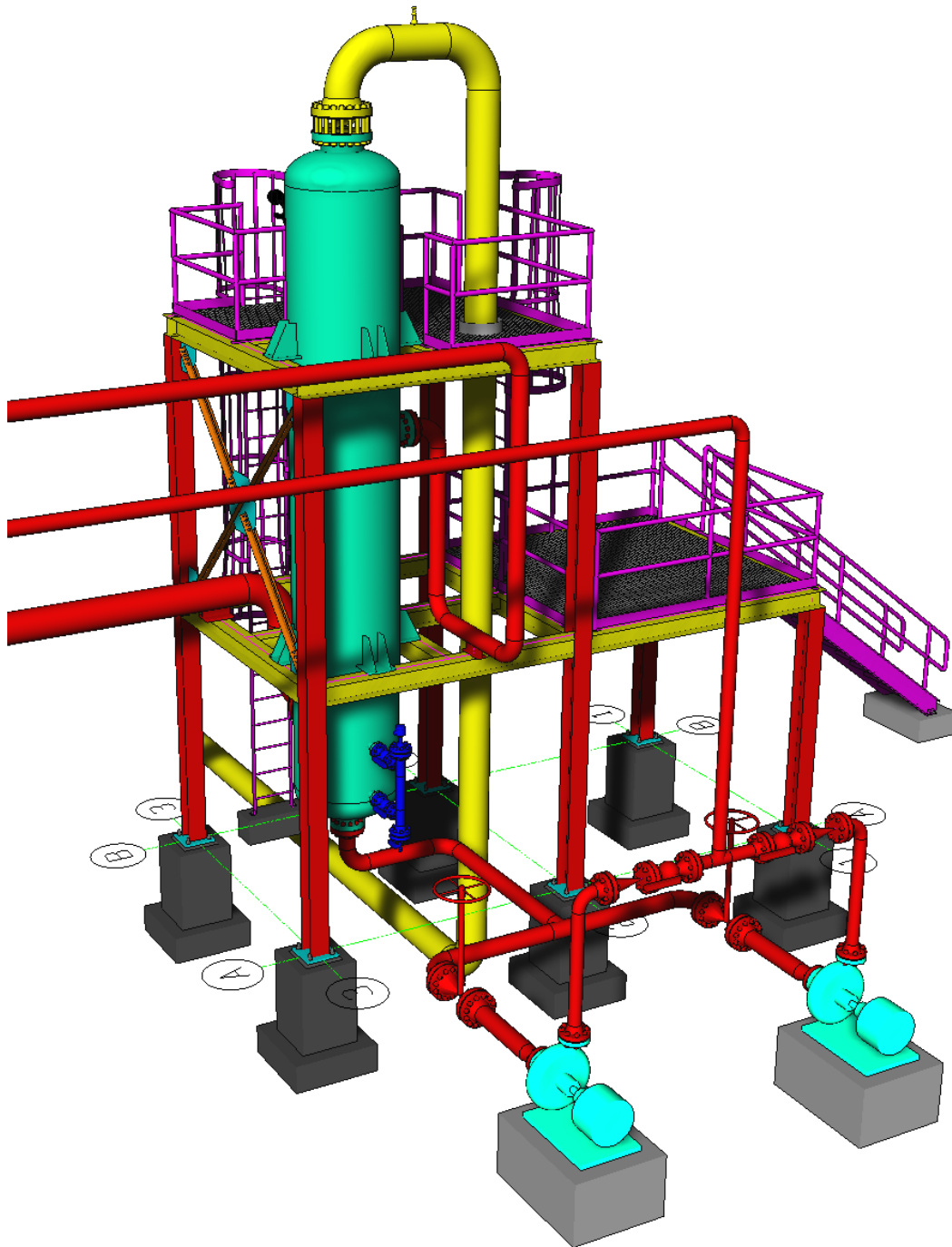




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Ammonia Scrubber

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2023

Analysis Performed by : SL User

Job File : ...\\Process Cond StripperOpR1.pvdb

Date of Analysis : Dec 2,2025 9:23am

PV Elite 27 U2, August 2025



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2023

Diameter Spec : 2750.000 mm ID	
Vessel Design Length, Tangent to Tangent	19820.00 mm
Distance of Bottom Tangent above Grade	6400.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	6100.00 mm
Internal Design Temperature	430 °C
Internal Design Pressure	5929.490 kPa
External Design Temperature	200 °C
External Design Pressure	103.421 kPa
Maximum Allowable Working Pressure	6009 kPa
External Max. Allowable Working Pressure	1408 kPa
Shop Test Pressure	8054.110 kPa
Field Test Pressure	7864.782 kPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	38.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normalized	Impact Tested
Shell	SA-387 11	2	...	K11789	No	Yes
Head	SA-387 11	2	...	K11789	No	Yes
Skirt	SA-516 70	...	...	K02700	No	No
Skirt	SA-387 11	...	...	K11789	No	No
Nozzle	SA-336 F11	3	...	K11572	No	Yes
Nozzle	SA-387 11	...	...	K11789	No	No
Nozzle	SA-335 P11	...	...	K11597	No	No
Re-Pad	SA-387 11	2	...	K11789	No	No
Nozzle Flg	SA-182 F11	2	...	K11572	No	No
Basering	SA-516 70	...	...	K02700	No	No
Base Bolting	SA-36	...	...	K02600	No	No

- Normalized is determined based on the UCS-66 material curve selection and Figure UCS-66.
- Impact Tested is based on material selection and material data properties.

Element Pressures and MAWP (kPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
K-3	...	...	...	1.6000	N/A	No
K-2	...	...	...	1.6000	N/A	No
K-1	...	...	...	1.6000	N/A	No
H-1	5958.832	103.42	6094.000	1.0000	No	No
S-1	5953.265	103.42	6121.000	1.0000	N/A	No
S-2	5930.622	103.42	6144.000	1.0000	N/A	No
S-3	5929.490	103.42	6145.000	1.0000	N/A	No
S-4	5929.490	103.42	6145.000	1.0000	N/A	No
S-5	5929.490	103.42	6145.000	1.0000	N/A	No
S-6	5929.490	103.42	6145.000	1.0000	N/A	No



S-7	5929.490	103.42	6145.000	1.0000	N/A	No
H-2	5929.490	103.42	6070.000	1.0000	No	No

Liquid Level: 3887.50 mm Dens.: 769.647 kg/m³ Sp. Gr.: 0.770

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-3600.0	2500.0	18.0	18.0	...	...	0.70	0.70
Skirt	-600.0	3000.0	18.0	18.0	...	...	0.70	0.70
Skirt	0.0	600.0	18.0	18.0	...	...	0.70	0.50
Ellipse	50.0	50.0	68.0	58.5	57.0	8.5	1.00	1.00
Cylinder	3050.0	3000.0	60.0	60.0	58.1	22.4	1.00	1.00
Cylinder	6050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	9050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	12050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	15050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	18050.0	3000.0	60.0	60.0	57.9	22.4	1.00	1.00
Cylinder	19770.0	1720.0	60.0	60.0	57.9	22.4	1.00	1.00
Ellipse	19820.0	50.0	68.0	58.0	56.7	8.5	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D-10"	WRC-107/537	0.873	n/a	Passed
E-8"	WRC-297	0.964	Shell	Passed
B-14"	WRC-297	0.961	Shell	Passed
A-10"	WRC-297	0.961	Shell	Passed
C-14"	WRC-107/537	0.876	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	6534.29	20387.1	95204.5	383837
20	30	4000	5967.56	20325.9	79577.2	332945
30	40	5800	5526.98	20128.7	62335.3	272263
40	50	6125	5438.87	20071.9	59045.6	260203
50	60	7650	5313.34	19779.7	58776.8	259207
60	70	10650	4598.79	18314.5	43908.6	202066
70	80	13650	3833.02	16872	31260.8	149286
80	90	16650	3049.32	15098.5	20937.3	101330
90	100	19650	2543.92	11617.2	12547.4	61256.3
100	110	22650	1792.24	8893.53	6043.18	30490.1
110	120	25010	1027.82	5316.59	1813.09	9174.89
120	130	25895	738.058	3655.96	294.437	1458.49

Abs Max of the all of the Stress Ratio's : 0.6538

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	44.0000	mm
Inside Diameter of Basing	2462.0000	mm
Outside Diameter of Basing	2982.0000	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	2814.0000	mm
Number of Bolts	20	

Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	158.0000	mm
Height of Gussets	274.0000	mm



Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	168.0000	mm

Basering Sketch for Cylindrical Skirts:

Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	10142.	kgf
Total Earthquake Shear on Support	23995.	kgf
Wind Moment on Support	244954.	kgf-m.
Earthquake Moment on Support	533586.	kgf-m.
Maximum Initial Bolt Load	26444.	kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	135984.6	kg
Shop Test - Fabricated + water (full)	264189.2	kg
Shipping - Fab. + rem. intls.+ shipping app.	174154.0	kg
Erected - Fab. + rem. intls.+ insul. (etc)	179814.0	kg
Empty - Fab. + intls. + details + wghts.	187957.2	kg
Operating - Empty + operating liquid (no ca)	231979.2	kg
Field Test - Empty weight + test medium (full)	302358.6	kg

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Nozzle Calculation Summary (kPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D-10"	6094	OK	...	OK	8.10	OK	Passed
MH-3-24"	6080	OK	...	...		OK	Passed
E-8"	6009	OK	...	OK	7.16	OK	Passed
UC-2-14"	6129	OK	...	...		OK	Passed
B-14"	6129	OK	...	OK	8.33	OK	Passed
MH-2-24"	6093	OK	...	...		OK	Passed
UC-1-14"	6129	OK	...	...		OK	Passed
A-10"	6075	OK	...	OK	8.10	OK	Passed
MH-1-24"	6093	OK	...	...		OK	Passed
C-14"	6070	OK	...	OK	8.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6009 Nozzle : E-8"
Minimum MAWP Shells/Flanges : 6009 Element : H-2
Minimum MAPnc Shells/Flanges : 6216 Element : H-2

Computed Vessel M.A.W.P. : 6009 kPa

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
40	D-10"	0.000	0.000	493.293
50	MH-3-24"	1550.000	180.000	1099.200
50	E-8"	550.000	0.000	357.607
60	UC-2-14"	5250.000	0.000	639.200
60	B-14"	3650.000	0.000	639.200
80	MH-2-24"	10550.000	0.000	1099.200
90	UC-1-14"	12550.001	0.000	639.200
100	A-10"	16900.004	0.000	446.087
110	MH-1-24"	18910.004	0.000	1099.200
120	C-14"	0.000	0.000	647.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
E-8"	8.000 in	160	WNF	8.625	23.012	819.08	62.00	564.37	600
D-10"	10.000 in	80	WNF	10.750	15.088	...	...	312.20	600
A-10"	10.000 in	160	WNF	10.750	28.575	593.05	62.00	566.79	600
UC-2-14"	14.000 in Actual		WNF	14.000	18.000	655.60	58.00	571.54	600
B-14"	14.000 in Actual		WNF	14.000	18.000	655.60	58.00	571.54	600
UC-1-14"	14.000 in Actual		WNF	14.000	18.000	655.60	58.00	571.54	600
C-14"	14.000 in Actual		WNF	14.000	16.000	655.60	58.00	564.28	900
MH-3-24"	24.000 in Actual		WNF	24.000	30.000	1049.60	58.00	594.21	600
MH-2-24"	24.000 in Actual		WNF	24.000	30.000	1049.60	58.00	594.21	600
MH-1-24"	24.000 in Actual		WNF	24.000	30.000	1049.60	58.00	594.21	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
E-8"	SA-335 P11	60.000	20.000	60.000	60.000	...
D-10"	SA-336 F11	58.500	10.000	...	...	...
A-10"	SA-335 P11	60.000	20.000	46.000	60.000	...
UC-2-14"	SA-387 11	58.000	10.000	44.000	58.000	...
B-14"	SA-387 11	58.000	10.000	44.000	58.000	...
UC-1-14"	SA-387 11	58.000	10.000	44.000	58.000	...
C-14"	SA-387 11	58.000	10.000	44.000	58.000	...
MH-3-24"	SA-387 11	58.000	10.000	44.000	58.000	...
MH-2-24"	SA-387 11	58.000	10.000	44.000	58.000	...
MH-1-24"	SA-387 11	58.000	10.000	44.000	58.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
E-8"	550.000	0.0	500.00	0.00	S-1
D-10"	...	0.0	250.00	0.00	H-1
A-10"	16900.002	0.0	500.00	0.00	S-6
UC-2-14"	5250.000	0.0	500.00	0.00	S-2
B-14"	3650.000	0.0	500.00	0.00	S-2
UC-1-14"	12550.001	0.0	500.00	0.00	S-5
C-14"	...	0.0	500.00	0.00	H-2
MH-3-24"	1550.000	180.0	500.00	0.00	S-1
MH-2-24"	10550.000	0.0	500.00	0.00	S-4
MH-1-24"	18910.004	0.0	500.00	0.00	S-7



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 68.0mm THK X 2750.0mm ID X 50.0mm	SA-387 11
6	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 2000.0mm	...
1	INSULATION: 2500mm X 50mm THK	
1	LINING: 2500.000mm X 50.000mm THK	...
1	INSULATION: 3000mm X 50mm THK	
1	LINING: 3000.000mm X 50.000mm THK	...
1	INSULATION: 600mm X 50mm THK	
1	LINING: 600.000mm X 50.000mm THK	...
2	INSULATION: 737mm X 140mm THK	hot
2	LINING: 737.500mm X 3.000mm THK	...
6	LADDER: 4000.0mm	...
6	INSULATION: 3000mm X 140mm THK	hot
6	LINING: 3000.000mm X 3.000mm THK	...
1	PACKING: 1175.000mm	...
2	PACKING: 3000.000mm	...
1	PACKING: 325.001mm	...
1	PACKING: 845.000mm	...
1	PACKING: 655.001mm	...
1	INSULATION: 1720mm X 140mm THK	hot
1	LINING: 1720.000mm X 3.000mm THK	...
1	PLATFORM: 1000mm X 6999mm WIDE	
3	CLASS 600 GR 1.9, 24.0" BLIND FLANGE(S)	SA-182 F11
2	CLASS 600 GR 1.9, 14.0" BLIND FLANGE(S)	SA-182 F11
1	BASERING: 2982.0mm OD X 2462.0 ID X 44.0mm	SA-516 70
20	BASERING BOLTS: 48.0mm DIA	SA-36
20	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 2982.0mm OD X 2646.0 ID X 44.0mm	...
40	BASERING GUSSET PLATES: 158.0mm X 158.0mm X 274.0mm X 16.0mm	...
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3006.5mm X 2500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2186.5mm X 2500.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 3006.5mm X 3000.0mm	SA-516 70



1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2186.5mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2892.8mm X 600.0mm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 18.0mm THK X 2413.9mm X 600.0mm	SA-387 11
9	CYLINDRICAL SEGMENT 60.0mm THK X 3048.0mm X 3000.0mm	SA-387 11
3	CYLINDRICAL SEGMENT 60.0mm THK X 2543.3mm X 3000.0mm	SA-387 11
3	CYLINDRICAL SEGMENT 60.0mm THK X 5591.3mm X 3000.0mm	SA-387 11
3	CYLINDRICAL SEGMENT 60.0mm THK X 0.0mm X 3000.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 60.0mm THK X 5591.3mm X 1720.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 60.0mm THK X 3048.0mm X 1720.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 60.0mm THK X 0.0mm X 1720.0mm	SA-387 11
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
H-1	[10]	!	-12	-13		0.980	58.500	1.00	Yes
H-1	[7]	!	-12	-20		0.859	68.000	1.00	Yes
S-1	[8]	!	-12	-13		0.975	60.000	1.00	Yes
S-2	[8]	!	-12	-14		0.971	60.000	1.00	Yes
S-3	[8]	!	-12	-14		0.971	60.000	1.00	Yes
S-4	[8]	!	-12	-14		0.971	60.000	1.00	Yes
S-5	[8]	!	-12	-14		0.971	60.000	1.00	Yes
S-6	[8]	!	-12	-14		0.971	60.000	1.00	Yes
S-7	[8]	!	-12	-14		0.971	60.000	1.00	Yes
H-2	[10]	!	-12	-13		0.983	58.000	1.00	Yes
H-2	[7]	!	-12	-20		0.855	68.000	1.00	Yes
D-10"	[1]	!	-12	-19		0.870	58.500	1.00	Yes
Nozzle Flg	[5]	B	-18	-42					
MH-3-24"	[1]	A	39	37		0.960	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
E-8"	[1]	A	40	38		0.961	60.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-42					
UC-2-14"	[1]	A	39	37		0.957	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
B-14"	[1]	A	39	37		0.957	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
MH-2-24"	[1]	A	39	37		0.957	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
UC-1-14"	[1]	A	39	37		0.957	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
A-10"	[1]	A	40	38		0.957	60.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-42					
MH-1-24"	[1]	A	39	37		0.957	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
C-14"	[1]	A	39	32		0.873	58.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No
Warmest MDMT:			40	38					
Required Minimum Design Metal Temperature						-12.0	°C		
Warmest Computed Minimum Design Metal Temperature						38.0	°C		

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld



[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 30 50 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (QEUP):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	kPa
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Ammonia Scrubber

Design Internal Pressure (for Hydrotest)	5929.5	kPa
Design Internal Temperature	430.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	Amine	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.000
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes



Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/12
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	K-3
Distance "FROM" to "TO"	2500 mm
Skirt Outside Diameter	2646 mm
Diameter of Skirt at Base	2646 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2386 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2473.7 kgf/cm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No
Element From Node	10
Detail Type	Platform
Detail ID	PL-6
Dist. from "FROM" Node / Offset dist	2100 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	90.0
Platform Wind Area	15000 cm ²
Platform Weight	982.15 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	175 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	2000 mm
Unit Weight of Ladder	0.037 kgf/mm
Platform Length (top head platform)	-1.0101 mm
Element From Node	10
Detail Type	Insulation
Detail ID	FR-3
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2500 mm
Thickness of Insulation	50 mm
Density	2400 kg/m ³
Element From Node	10
Detail Type	Lining
Detail ID	FR-3.1
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	2500 mm
Thickness of Lining	50 mm
Density	2400 kg/m ³



Element From Node	10
Detail Type	Weight
Detail ID	Taling Lug
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	1000 kgf
Offset from Element Centerline	0 mm

Element#	2/12
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	K-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	2646 mm
Diameter of Skirt at Base	2646 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element From Node	20
Detail Type	Insulation
Detail ID	FR-2
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	50 mm
Density	2400 kg/m ³

Element From Node	20
Detail Type	Lining
Detail ID	FR-2.1
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	50 mm
Density	2400 kg/m ³

Element From Node	20
Detail Type	Weight
Detail ID	Skirt MW
Dist. from "FROM" Node / Offset dist	0.0002896 mm
Miscellaneous Weight	500 kgf
Offset from Element Centerline	0 mm

Element#	3/12
Element From Node	30
Element To Node	40
Element Type	Skirt Support
Description	K-1
Distance "FROM" to "TO"	600 mm
Skirt Outside Diameter	2646 mm
Diameter of Skirt at Base	2646 mm
Skirt Thickness	18 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	200 °C
Design Temperature External Pressure	200 °C



Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	1509.1 kgf/cm ²
Allowable Stress, Operating	1509.1 kgf/cm ²
Allowable Stress, Hydrotest	2844.9 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	2671.6 kgf/cm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element From Node	30
Detail Type	Insulation
Detail ID	FR-1
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	600 mm
Thickness of Insulation	50 mm
Density	2400 kg/m ³

Element From Node	30
Detail Type	Lining
Detail ID	FR-1.1
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	600 mm
Thickness of Lining	50 mm
Density	2400 kg/m ³

Element#	4/12
Element From Node	40
Element To Node	50
Element Type	Elliptical
Description	H-1
Distance "FROM" to "TO"	50 mm
Inside Diameter	2750 mm
Element Thickness	58.5 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5929.5 kPa
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 kPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 [Impact Tested]
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	-687.5 mm
Height/Length of Liquid	737.5 mm
Liquid Density	769.65 kg/m ³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-687.5 mm
Height/Length of Insulation	737.5 mm



Thickness of Insulation	140	mm
Density	352	kg/m ³
Element From Node	40	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	-687.5	mm
Height/Length of Lining	737.5	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	D-10"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	466.92	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	[Impact Tested]
Element From Node	40	
Detail Type	Weight	
Detail ID	PPD	
Dist. from "FROM" Node / Offset dist	-685	mm
Miscellaneous Weight	246	kgf
Offset from Element Centerline	2840	mm

Element#	5/12	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	S-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	[Impact Tested]
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PL-1	
Dist. from "FROM" Node / Offset dist	0.48768E+04	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	30000	cm ²
Platform Weight	3931.3	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	

Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	769.65	kg/m ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	140	mm
Density	352	kg/m ³

Element From Node	50	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	MH-3-24"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1453.4	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-387 11	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	E-8"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	8	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	389.11	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-335 P11	

Element From Node	50	
Detail Type	Weight	
Detail ID	PPE	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	1818	kgf
Offset from Element Centerline	2815	mm

Element#	6/12	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	S-2	
Distance "FROM" to "TO"	3000	mm



Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	[Impact Tested]
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Platform	
Detail ID	PL-2	
Dist. from "FROM" Node / Offset dist	1610	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	30000	cm ²
Platform Weight	2039.6	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	60	
Detail Type	Packing	
Detail ID	Bed#2	
Dist. from "FROM" Node / Offset dist	1825	mm
Height of Packed Section	1175	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	150	mm
Liquid Density	769.65	kg/m ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	140	mm
Density	352	kg/m ³

Element From Node	60	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	60	
Detail Type	Nozzle	



Detail ID UC-2-14"
Dist. from "FROM" Node / Offset dist 2200 mm
Nozzle Diameter 14 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 0.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 479.06 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-387 11

Element From Node 60
Detail Type Nozzle
Detail ID B-14"
Dist. from "FROM" Node / Offset dist 600 mm
Nozzle Diameter 14 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 0.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 324.33 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-387 11

Element From Node 60
Detail Type Weight
Detail ID DLBS-1
Dist. from "FROM" Node / Offset dist 1825 mm
Miscellaneous Weight 630 kgf
Offset from Element Centerline 0 mm

Element From Node 60
Detail Type Weight
Detail ID PPB
Dist. from "FROM" Node / Offset dist 600 mm
Miscellaneous Weight 3286 kgf
Offset from Element Centerline 2891 mm

Element# 7/12
Element From Node 70
Element To Node 80
Element Type Cylinder
Description S-3
Distance "FROM" to "TO" 3000 mm
Inside Diameter 2750 mm
Element Thickness 60 mm
Internal Corrosion Allowance 1 mm
Nominal Thickness 60 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 5929.5 kPa
Design Temperature Internal Pressure 430 °C
Design External Pressure 103.42 kPa
Design Temperature External Pressure 200 °C
Effective Diameter Multiplier 1.2
Material Name SA-387 11 [Impact Tested]
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Weld is pre-Heated No

Element From Node 70
Detail Type Platform
Detail ID PL-3
Dist. from "FROM" Node / Offset dist 2610 mm
Platform Start Angle (degrees) 0.0
Platform End Angle (degrees) 180.0
Platform Wind Area 30000 cm²
Platform Weight 2039.6 kgf

Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	70	
Detail Type	Packing	
Detail ID	Bed#2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	140	mm
Density	352	kg/m ³

Element From Node	70	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element#	8/12	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	S-4	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	[Impact Tested]
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	Bed#2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	325	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	80
Detail Type	Packing
Detail ID	Bed#1
Dist. from "FROM" Node / Offset dist	2155 mm
Height of Packed Section	845 mm
Density	153 kg/m ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	0.76999998

Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	140 mm
Density	352 kg/m ³

Element From Node	80
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³

Element From Node	80
Detail Type	Nozzle
Detail ID	MH-2-24"
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1453.4 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-387 11

Element From Node	80
Detail Type	Weight
Detail ID	DLBS-2
Dist. from "FROM" Node / Offset dist	2155 mm
Miscellaneous Weight	630 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	DLLD-2
Dist. from "FROM" Node / Offset dist	800 mm
Miscellaneous Weight	800 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	DLBL-2
Dist. from "FROM" Node / Offset dist	325 mm
Miscellaneous Weight	160 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	OPLD-2
Dist. from "FROM" Node / Offset dist	800 mm
Miscellaneous Weight	1100 kgf
Offset from Element Centerline	0 mm

Element From Node	80
Detail Type	Weight
Detail ID	TALD-2



Dist. from "FROM" Node / Offset dist	770	mm
Miscellaneous Weight	100	kgf
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	small Noz	
Dist. from "FROM" Node / Offset dist	950	mm
Miscellaneous Weight	160	kgf
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Cleat Weight	
Dist. from "FROM" Node / Offset dist	950	mm
Miscellaneous Weight	4252	kgf
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Margin Weight	
Dist. from "FROM" Node / Offset dist	950	mm
Miscellaneous Weight	10000	kgf
Offset from Element Centerline	0	mm

Element#	9/12	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	S-5	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2750	mm
Element Thickness	60	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	60	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5929.5	kPa
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	kPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	[Impact Tested]
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Platform	
Detail ID	PL-4	
Dist. from "FROM" Node / Offset dist	610	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	30000	cm ²
Platform Weight	2039.6	kgf
Platform Railing Weight	0	kgf/mm
Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	90	
-------------------	----	--

Detail Type	Packing
Detail ID	Bed#1
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	3000 mm
Density	153 kg/m ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	0.76999998

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	140 mm
Density	352 kg/m ³

Element From Node	90
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	8027 kg/m ³

Element From Node	90
Detail Type	Nozzle
Detail ID	UC-1-14"
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	14 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	479.06 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-387 11

Element#	10/12
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	S-6
Distance "FROM" to "TO"	3000 mm
Inside Diameter	2750 mm
Element Thickness	60 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5929.5 kPa
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 kPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 [Impact Tested]
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	100
Detail Type	Platform
Detail ID	PL-5
Dist. from "FROM" Node / Offset dist	1610 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	30000 cm ²
Platform Weight	2039.6 kgf
Platform Railing Weight	0 kgf/mm

Platform Grating Weight	1668	N/m ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	175	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	-1.0101	mm

Element From Node	100	
Detail Type	Packing	
Detail ID	Bed#1	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	655	mm
Density	153	kg/m ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	0.76999998	

Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	140	mm
Density	352	kg/m ³

Element From Node	100	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	100	
Detail Type	Nozzle	
Detail ID	A-10"	
Dist. from "FROM" Node / Offset dist	1850	mm
Nozzle Diameter	10	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	311.06	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-335 P11	

Element From Node	100	
Detail Type	Weight	
Detail ID	DLLD-1	
Dist. from "FROM" Node / Offset dist	1110	mm
Miscellaneous Weight	800	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	DLBL-1	
Dist. from "FROM" Node / Offset dist	655	mm
Miscellaneous Weight	160	kgf
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	DLFP	
Dist. from "FROM" Node / Offset dist	1850	mm
Miscellaneous Weight	250	kgf
Offset from Element Centerline	0	mm



Element From Node 100
Detail Type Weight
Detail ID OPLD-1
Dist. from "FROM" Node / Offset dist 1110 mm
Miscellaneous Weight 1100 kgf
Offset from Element Centerline 0 mm

Element From Node 100
Detail Type Weight
Detail ID OPFP
Dist. from "FROM" Node / Offset dist 1850 mm
Miscellaneous Weight 140 kgf
Offset from Element Centerline 0 mm

Element From Node 100
Detail Type Weight
Detail ID TALD-1
Dist. from "FROM" Node / Offset dist 1200 mm
Miscellaneous Weight 100 kgf
Offset from Element Centerline 0 mm

Element From Node 100
Detail Type Weight
Detail ID PPA
Dist. from "FROM" Node / Offset dist 1850 mm
Miscellaneous Weight 5182 kgf
Offset from Element Centerline 2840 mm

Element# 11/12
Element From Node 110
Element To Node 120
Element Type Cylinder
Description S-7
Distance "FROM" to "TO" 1720 mm
Inside Diameter 2750 mm
Element Thickness 60 mm
Internal Corrosion Allowance 1 mm
Nominal Thickness 60 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 5929.5 kPa
Design Temperature Internal Pressure 430 °C
Design External Pressure 103.42 kPa
Design Temperature External Pressure 200 °C
Effective Diameter Multiplier 1.2
Material Name SA-387 11 [Impact Tested]
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Weld is pre-Heated No

Element From Node 110
Detail Type Insulation
Detail ID Ins: 80
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Insulation 1720 mm
Thickness of Insulation 140 mm
Density 352 kg/m³

Element From Node 110
Detail Type Lining
Detail ID clad
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Lining 1720 mm
Thickness of Lining 3 mm
Density 8027 kg/m³

Element From Node 110



Detail Type	Nozzle
Detail ID	MH-1-24"
Dist. from "FROM" Node / Offset dist	860 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1453.4 kgf
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-387 11

Element From Node	110
Detail Type	Weight
Detail ID	DLPAD
Dist. from "FROM" Node / Offset dist	1350 mm
Miscellaneous Weight	230 kgf
Offset from Element Centerline	0 mm

Element From Node	110
Detail Type	Weight
Detail ID	TAPAD
Dist. from "FROM" Node / Offset dist	1350 mm
Miscellaneous Weight	200 kgf
Offset from Element Centerline	0 mm

Element From Node	110
Detail Type	Weight
Detail ID	Pipe Davit
Dist. from "FROM" Node / Offset dist	1720 mm
Miscellaneous Weight	600 kgf
Offset from Element Centerline	1400 mm

Element#	12/12
Element From Node	120
Element To Node	130
Element Type	Elliptical
Description	H-2
Distance "FROM" to "TO"	50 mm
Inside Diameter	2750 mm
Element Thickness	58 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5929.5 kPa
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 kPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 [Impact Tested]
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	120
Detail Type	Platform
Detail ID	TOP PL
Dist. from "FROM" Node / Offset dist	890 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	70000 cm ²
Platform Weight	8482.6 kgf
Platform Railing Weight	0 kgf/mm
Platform Grating Weight	1668 N/m ²
Platform Width	7000 mm
Platform Height	1000 mm



Platform Clearance or End Offset	3500	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.037	kgf/mm
Platform Length (top head platform)	7000	mm

Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	737.5	mm
Thickness of Insulation	140	mm
Density	352	kg/m ³

Element From Node	120	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	737.5	mm
Thickness of Lining	3	mm
Density	8027	kg/m ³

Element From Node	120	
Detail Type	Nozzle	
Detail ID	C-14"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	14	in.
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	377.95	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-387 11	

Element From Node	120	
Detail Type	Weight	
Detail ID	PPC	
Dist. from "FROM" Node / Offset dist	687	mm
Miscellaneous Weight	4218	kgf
Offset from Element Centerline	2891	mm

Element From Node	120	
Detail Type	Weight	
Detail ID	PSV	
Dist. from "FROM" Node / Offset dist	642	mm
Miscellaneous Weight	2000	kgf
Offset from Element Centerline	3500	mm

Element From Node	120	
Detail Type	Weight	
Detail ID	Lifting Lug	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	1500	kgf
Offset from Element Centerline	0	mm

Element From Node	120	
Detail Type	For./Mom.	
Detail ID	CFxFZ	
Dist. from "FROM" Node / Offset dist	685	mm
Force in X Direction	2550.9	kgf
Force in Y Direction	-3532.1	kgf
Force in Z Direction	2550.9	kgf
Moment about X Axis	2746.7	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	2746.7	kgf-m.
Force/Moment Combination Method	SRSS	

Ammonia Scrubber_
PV Elite 27 U2 Licensee: SL User
FileName : Process Cond Stripper_Op_R1
Input Echo:

Step: 1 9:23am Dec 2, 2025

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	K-3	...	2500	...	2500
	K-2	...	5500	...	3000
	K-1	...	6100	...	600
	H-1	...	6150	...	50
	S-1	...	9150	...	3000
	S-2	...	12150	...	3000
	S-3	...	15150	...	3000
	S-4	...	18150	...	3000
	S-5	...	21150	...	3000
	S-6	...	24150	...	3000
	S-7	...	25870	...	1720
	H-2	...	25920	...	50

PV Elite includes an 1/8 inch (3.175mm) raised face and gasket thicknesses for girth flanges and tubesheet thicknesses where applicable in the Tangent to Tangent length calculation. The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
	K-3	...	18	1.6	2646	...	1
	K-2	...	18	1.6	2646	...	2
	K-1	...	18	1.6	2646	...	3
	H-1	5958.8	68	1	2750	1499	4
	S-1	5953.3	60	1	2750	1499	5
	S-2	5930.6	60	1	2750	1499	6
	S-3	5929.5	60	1	2750	1499	7
	S-4	5929.5	60	1	2750	1499	8
	S-5	5929.5	60	1	2750	1499	9
	S-6	5929.5	60	1	2750	1499	10
	S-7	5929.5	60	1	2750	1499	11
	H-2	5929.5	68	1	2750	1499	12

Element Required Thickness and MAWP :

From	To	Design Pressure kPa	M.A.W.P. Corroded kPa	M.A.P. New & Cold kPa	Minimum Thickness mm	Required Thickness mm	Element#
	K-3	...	No Calc	No Calc	18	No Calc	1
	K-2	...	No Calc	No Calc	18	No Calc	2
	K-1	...	No Calc	No Calc	18	No Calc	3
	H-1	5929.49	6093.65	6269.57	58.5	56.9522	4
	S-1	5929.49	6121.03	6292.93	60	58.1151	5
	S-2	5929.49	6143.67	6292.93	60	57.8925	6
	S-3	5929.49	6144.81	6292.93	60	57.8813	7
	S-4	5929.49	6144.81	6292.93	60	57.8813	8
	S-5	5929.49	6144.81	6292.93	60	57.8813	9
	S-6	5929.49	6144.81	6292.93	60	57.8813	10
	S-7	5929.49	6144.81	6292.93	60	57.8813	11
	H-2	5929.49	6069.96	6216.21	58	56.6755	12

Minimum 6009 6216

MAWP: 6009 kPa, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2023

Internal Pressure Results for: H-1

Elliptical Head Design Information:					
Design Pressure	P	5958.832 kPa	Int. Design Temperature		430 °C
Material		SA-387 11	Ext. Design Temperature		200 °C
Length		50.000 mm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	2750.000 mm	Actual Stress		1458.8 kgf/cm ²
Spec. Min. Thk t _e , or s _a		58.5000 mm	Spec. Nominal Thk.		68.000 mm
Surface Area		94818.4 cm ²	SG of Contents		0.770
Element Volume		3019284.5 cm ³	Weight of Contents		2309.1 kgf
Empty Weight		5242.3 kgf	Operating Weight		7551.4 kgf
			Comp. All. Ope. Stress		823.8 kgf/cm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5958.832 \cdot 2752.0005 \cdot 0.999) / (2 \cdot 1498.98 \cdot 1.0 - 0.2 \cdot 5958.832)$$

$$= 55.9522 + 1.0000 = 56.9522 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 29.341 kPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1498.98 \cdot 1.0 \cdot 57.5) / (0.999 \cdot 2752.0005 + 0.2 \cdot 57.5)$$

$$= 6123 - 29 = 6094 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 58.5) / (1.0 \cdot 2750.0002 + 0.2 \cdot 58.5)$$

$$= 6270 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5958.832 \cdot (0.999 \cdot 2752.0005 + 0.2 \cdot 57.5)) / (2 \cdot 1.0 \cdot 57.5)$$

$$= 1458.792 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5958.832 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5958.832) + 1.0$$

$$= 58.170 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 24.152 kPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1498.98 \cdot 1.0 \cdot 67.0) / (1376.0002 + 0.6 \cdot 67.0)$$

$$= 6954 - 24 = 6930 \text{ kPa}$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2752.0 / (2 \cdot 688.5))^2) / 6$$

$$= 0.999032$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 10.169 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

EXTERNAL PRESSURE RESULTS for: H-1

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
57.500	2867.00	49.86	0.0027855	1002.40

$$MAEP = B / (K0 \cdot Do / t) = 1002.3997 / (0.9 \cdot 49.8609) = 2190.5291 \text{ kPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.544	2867.00	380.05	0.0003655	360.75

$$MAEP = B / (K0 \cdot Do / t) = 360.7451 / (0.9 \cdot 380.0473) = 103.4262 \text{ kPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2023

Internal Pressure Results for: S-1

Cylindrical Shell Design Information:					
Design Pressure	P	5953.265 kPa	Int. Design Temperature		430 °C
Material		SA-387 11	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress		1452.3 kgf/cm²
Spec. Min. Thk t,e, or sa		60.0000 mm	Spec. Nominal Thk.		60.000 mm
Surface Area		270491.2 cm²	SG of Contents		0.770
Element Volume		17818726.0 cm³	Weight of Contents		13654.5 kgf
Empty Weight		12315.5 kgf	Operating Weight		25970.0 kgf
			Comp. All. Ope. Stress		827.0 kgf/cm²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Efficiency	E	100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5953.265 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5953.265)$
 $= 57.1151 + 1.0000 = 58.1151 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
[Less Operating Hydrostatic Head Pressure of 23.775 kPa](#)
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1498.98 \cdot 1.0 \cdot 59.0) / (1376.0002 + 0.6 \cdot 59.0)$
 $= 6145 - 24 = 6121 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509.1 \cdot 1.0 \cdot 60.0) / (1375.0 + 0.6 \cdot 60.0)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5953.265 \cdot (1376.0002 + 0.6 \cdot 59.0)) / (1.0 \cdot 59.0)$
 $= 1452.255 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: S-1

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²



Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*523.6461)/(3*48.6441) = 1407.5282 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.9126)/(3*133.898) = 103.4243 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463.2263)/(3*48.6441) = 1245.1238 \text{ kPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2023

Internal Pressure Results for: S-2

Cylindrical Shell Design Information:					
Design Pressure	P	5930.622 kPa	Int. Design Temperature		430 °C
Material		SA-387 11	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress		1446.7 kgf/cm²
Spec. Min. Thk t _e , or s _a		60.0000 mm	Spec. Nominal Thk.		60.000 mm
Surface Area		270491.2 cm²	SG of Contents		0.770
Element Volume		17818726.0 cm³	Weight of Contents		682.7 kgf
Empty Weight		12315.5 kgf	Operating Weight		12998.2 kgf
			Comp. All. Ope. Stress		827.0 kgf/cm²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Efficiency	E	100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5930.622 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5930.622)$
 $= 56.8925 + 1.0000 = 57.8925 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
Less Operating Hydrostatic Head Pressure of 1.132 kPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1498.98 \cdot 1.0 \cdot 59.0) / (1376.0002 + 0.6 \cdot 59.0)$
 $= 6145 - 1 = 6144 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509.1 \cdot 1.0 \cdot 60.0) / (1375.0 + 0.6 \cdot 60.0)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5930.622 \cdot (1376.0002 + 0.6 \cdot 59.0)) / (1.0 \cdot 59.0)$
 $= 1446.731 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: S-2

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²



Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*523.6461)/(3*48.6441) = 1407.5282 \text{ kPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.9126)/(3*133.898) = 103.4243 \text{ kPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463.2263)/(3*48.6441) = 1245.1238 \text{ kPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: S-3

Cylindrical Shell Design Information:					
Design Pressure	P	5929.490 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1499.0 kgf/cm²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1509.1 kgf/cm²	
Inside Diameter	Di	2750.000 mm	Actual Stress	1446.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		60.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		270491.2 cm²	SG of Contents	0.000	
Element Volume		17818726.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		12315.5 kgf	Operating Weight	12315.5 kgf	
			Comp. All. Ope. Stress	827.0 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E	100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5929.49 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 + 1.0000 = 57.8813 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1498.98 \cdot 1.0 \cdot 59.0) / (1376.0002 + 0.6 \cdot 59.0)$
 $= 6145 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509.1 \cdot 1.0 \cdot 60.0) / (1375.0 + 0.6 \cdot 60.0)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5929.49 \cdot (1376.0002 + 0.6 \cdot 59.0)) / (1.0 \cdot 59.0)$
 $= 1446.455 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: S-3

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²



Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*523.6461)/(3*48.6441) = 1407.5282 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.9126)/(3*133.898) = 103.4243 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463.2263)/(3*48.6441) = 1245.1238 \text{ kPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: S-4

Cylindrical Shell Design Information:				
Design Pressure	P	5929.490 kPa	Int. Design Temperature	430 °C
Material		SA-387 11	Ext. Design Temperature	200 °C
Length		3000.000 mm	External Pressure Chart	CS-3
			UNS Number	K11789
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope) S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress	1446.5 kgf/cm²
Spec. Min. Thk t _e , or s _a		60.0000 mm	Spec. Nominal Thk.	60.000 mm
Surface Area		270491.2 cm²	SG of Contents	0.000
Element Volume		17818726.0 cm³	Weight of Contents	0.0 kgf
Empty Weight		12315.5 kgf	Operating Weight	12315.5 kgf
			Comp. All. Ope. Stress	827.0 kgf/cm²

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E	100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5929.49 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 + 1.0000 = 57.8813 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1498.98 \cdot 1.0 \cdot 59.0) / (1376.0002 + 0.6 \cdot 59.0)$
 $= 6145 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509.1 \cdot 1.0 \cdot 60.0) / (1375.0 + 0.6 \cdot 60.0)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5929.49 \cdot (1376.0002 + 0.6 \cdot 59.0)) / (1.0 \cdot 59.0)$
 $= 1446.455 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: S-4

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²



Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*523.6461)/(3*48.6441) = 1407.5282 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.9126)/(3*133.898) = 103.4243 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463.2263)/(3*48.6441) = 1245.1238 \text{ kPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: S-5

Cylindrical Shell Design Information:					
Design Pressure	P	5929.490 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	2750.000 mm	Actual Stress	1446.5 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		60.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		270491.2 cm ²	SG of Contents	0.000	
Element Volume		17818726.0 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		12315.5 kgf	Operating Weight	12315.5 kgf	
			Comp. All. Ope. Stress	827.0 kgf/cm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5929.49 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 + 1.0000 = 57.8813 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1498.98 \cdot 1.0 \cdot 59.0) / (1376.0002 + 0.6 \cdot 59.0)$
 $= 6145 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509.1 \cdot 1.0 \cdot 60.0) / (1375.0 + 0.6 \cdot 60.0)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5929.49 \cdot (1376.0002 + 0.6 \cdot 59.0)) / (1.0 \cdot 59.0)$
 $= 1446.455 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: S-5

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²



Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*523.6461)/(3*48.6441) = 1407.5282 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.9126)/(3*133.898) = 103.4243 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463.2263)/(3*48.6441) = 1245.1238 \text{ kPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: S-6

Cylindrical Shell Design Information:					
Design Pressure	P	5929.490 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	2750.000 mm	Actual Stress	1446.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		60.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		270491.2 cm²	SG of Contents	0.000	
Element Volume		17818726.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		12315.5 kgf	Operating Weight	12315.5 kgf	
			Comp. All. Ope. Stress	827.0 kgf/cm²	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E	100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5929.49 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 + 1.0000 = 57.8813 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1498.98 \cdot 1.0 \cdot 59.0) / (1376.0002 + 0.6 \cdot 59.0)$
 $= 6145 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509.1 \cdot 1.0 \cdot 60.0) / (1375.0 + 0.6 \cdot 60.0)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5929.49 \cdot (1376.0002 + 0.6 \cdot 59.0)) / (1.0 \cdot 59.0)$
 $= 1446.455 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: S-6

Cylindrical Shell From 100 to 110 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²



Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*523.6461)/(3*48.6441) = 1407.5282 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.9126)/(3*133.898) = 103.4243 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463.2263)/(3*48.6441) = 1245.1238 \text{ kPa}$$



Internal Pressure Calculation Results:

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Internal Pressure Results for: S-7

Cylindrical Shell Design Information:					
Design Pressure	P	5929.490 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		1720.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	2750.000 mm	Actual Stress	1446.5 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		60.0000 mm	Spec. Nominal Thk.	60.000 mm	
Surface Area		155081.6 cm ²	SG of Contents	0.000	
Element Volume		10216071.0 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		7060.9 kgf	Operating Weight	7060.9 kgf	
			Comp. All. Ope. Stress	827.0 kgf/cm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (5929.49 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 + 1.0000 = 57.8813 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1498.98 \cdot 1.0 \cdot 59.0) / (1376.0002 + 0.6 \cdot 59.0)$
 $= 6145 \text{ kPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1509.1 \cdot 1.0 \cdot 60.0) / (1375.0 + 0.6 \cdot 60.0)$
 $= 6293 \text{ kPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (5929.49 \cdot (1376.0002 + 0.6 \cdot 59.0)) / (1.0 \cdot 59.0)$
 $= 1446.455 \text{ kgf/cm}^2$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.135 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: S-7

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²



Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	20278.33	48.64	7.0656	0.0005305	523.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*523.6461)/(3*48.6441) = 1407.5282 \text{ kPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.434	2870.00	20278.33	133.90	7.0656	0.0001073	105.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.9126)/(3*133.898) = 103.4243 \text{ kPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
59.000	2870.00	902455.62	48.64	50.0000	0.0004693	463.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*463.2263)/(3*48.6441) = 1245.1238 \text{ kPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: H-2

Elliptical Head Design Information:					
Design Pressure	P	5929.490 kPa	Int. Design Temperature	430 °C	
Material		SA-387 11	Ext. Design Temperature	200 °C	
Length		50.000 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	1499.0 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm ²
Inside Diameter	Di	2750.000 mm	Actual Stress	1464.3 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		58.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		94818.4 cm ²	SG of Contents	0.000	
Element Volume		3019284.5 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		5242.3 kgf	Operating Weight	5242.3 kgf	
			Comp. All. Ope. Stress	822.7 kgf/cm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5929.49 \cdot 2752.0005 \cdot 0.999) / (2 \cdot 1498.98 \cdot 1.0 - 0.2 \cdot 5929.49)$$

$$= 55.6755 + 1.0000 = 56.6755 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1498.98 \cdot 1.0 \cdot 57.0) / (0.999 \cdot 2752.0005 + 0.2 \cdot 57.0)$$

$$= 6070 \text{ kPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 58.0) / (1.0 \cdot 2750.0002 + 0.2 \cdot 58.0)$$

$$= 6216 \text{ kPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5929.49 \cdot (0.999 \cdot 2752.0005 + 0.2 \cdot 57.0)) / (2 \cdot 1.0 \cdot 57.0)$$

$$= 1464.290 \text{ kgf/cm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5929.49 \cdot 1376.0002) / (1498.98 \cdot 1.0 - 0.6 \cdot 5929.49) + 1.0$$

$$= 57.881 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1498.98 \cdot 1.0 \cdot 67.0) / (1376.0002 + 0.6 \cdot 67.0)$$

$$= 6954 \text{ kPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2752.0 / (2 \cdot 688.5))^2) / 6$$

= 0.999032

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 10.169 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.375 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.188 mm

EXTERNAL PRESSURE RESULTS for: H-2

Elliptical Head From 120 to 130 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 1974242 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
57.000	2866.00	50.28	0.0027623	1000.96

$$MAEP = B / (K0 \cdot Do/t) = 1000.9638 / (0.9 \cdot 50.2807) = 2169.127 \text{ kPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.541	2866.00	380.05	0.0003655	360.75

$$MAEP = B / (K0 \cdot Do/t) = 360.745 / (0.9 \cdot 380.0474) = 103.4262 \text{ kPa}$$

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Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 7864.782 kPa
UG99b(b)(35) = 1.30 * Design Pres * Sa/S 7760.377 kPa
UG99(c) = 1.30 * M.A.P. - Head(Hyd) 8054.110 kPa
UG100 = 1.10 * M.A.W.P. * Sa/S 6654.815 kPa
PED = max(1.43*DP, 1.25*DP*ratio) 8479.170 kPa
PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 8479.170 kPa
App 27-4 = M.A.W.P. 6009.262 kPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm² & kPa):

From To	Stress	Allowable	Ratio	Pressure
H-1	1945.1	2844.9	0.684	8081.07
S-1	1937.9	2844.9	0.681	8081.07
S-2	1937.9	2844.9	0.681	8081.07
S-3	1937.9	2844.9	0.681	8081.07
S-4	1937.9	2844.9	0.681	8081.07
S-5	1937.9	2844.9	0.681	8081.07
S-6	1937.9	2844.9	0.681	8081.07
S-7	1937.9	2844.9	0.681	8081.07
H-2	1961.8	2844.9	0.690	8081.07

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D-10"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Pad	1509.10	1498.98	1.007
E-8"	Nozzle	1203.20	1009.30	1.192
E-8"	Pad	1509.10	1498.98	1.007
UC-2-14"	Nozzle	1509.10	1498.98	1.007
UC-2-14"	Pad	1509.10	1498.98	1.007
B-14"	Nozzle	1509.10	1498.98	1.007
B-14"	Pad	1509.10	1498.98	1.007
MH-2-24"	Nozzle	1509.10	1498.98	1.007
MH-2-24"	Pad	1509.10	1498.98	1.007
UC-1-14"	Nozzle	1509.10	1498.98	1.007
UC-1-14"	Pad	1509.10	1498.98	1.007
A-10"	Nozzle	1203.20	1009.30	1.192
A-10"	Pad	1509.10	1498.98	1.007
MH-1-24"	Nozzle	1509.10	1498.98	1.007
MH-1-24"	Pad	1509.10	1498.98	1.007
C-14"	Nozzle	1509.10	1498.98	1.007
C-14"	Pad	1509.10	1498.98	1.007

Minimum 1.007

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H-1	1509.10	1498.98	1.007
S-1	1509.10	1498.98	1.007
S-2	1509.10	1498.98	1.007
S-3	1509.10	1498.98	1.007
S-4	1509.10	1498.98	1.007
S-5	1509.10	1498.98	1.007
S-6	1509.10	1498.98	1.007



S-7	1509.10	1498.98	1.007
H-2	1509.10	1498.98	1.007

Minimum 1.007

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
D-10"	819.24	2844.86	0.288
MH-3-24"	804.28	2844.86	0.283
E-8"	415.32	1881.28	0.221
UC-2-14"	781.02	2844.86	0.275
B-14"	781.02	2844.86	0.275
MH-2-24"	804.28	2844.86	0.283
UC-1-14"	781.02	2844.86	0.275
A-10"	417.00	1881.28	0.222
MH-1-24"	804.28	2844.86	0.283
C-14"	882.77	2844.86	0.310

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7864.782 kPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7760.377 kPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	8073.836 kPa
UG100	= 1.10 * M.A.W.P. * Sa/S	6654.815 kPa
PED	= max(1.43*DP, 1.25*DP*ratio)	8479.170 kPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8479.170 kPa
App 27-4	= M.A.W.P.	6009.262 kPa

UG-99(b), Test Pressure Calculation:
= Test Factor • MAWP • Stress Ratio
= 1.3 • 6009.262 • 1.007
= 7864.782 kPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm² & kPa):

From To	Stress	Allowable	Ratio	Pressure
H-1	1976.3	2844.9	0.695	8072.54
S-1	1967.5	2844.9	0.692	8065.31
S-2	1960.3	2844.9	0.689	8035.90
S-3	1953.1	2844.9	0.687	8006.49
S-4	1946.0	2844.9	0.684	7977.09
S-5	1938.8	2844.9	0.682	7947.68
S-6	1931.6	2844.9	0.679	7918.28
S-7	1924.4	2844.9	0.676	7888.87
H-2	1944.0	2844.9	0.683	7872.01

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D-10"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Nozzle	1509.10	1498.98	1.007
MH-3-24"	Pad	1509.10	1498.98	1.007
E-8"	Nozzle	1203.20	1009.30	1.192
E-8"	Pad	1509.10	1498.98	1.007
UC-2-14"	Nozzle	1509.10	1498.98	1.007
UC-2-14"	Pad	1509.10	1498.98	1.007
B-14"	Nozzle	1509.10	1498.98	1.007
B-14"	Pad	1509.10	1498.98	1.007
MH-2-24"	Nozzle	1509.10	1498.98	1.007
MH-2-24"	Pad	1509.10	1498.98	1.007
UC-1-14"	Nozzle	1509.10	1498.98	1.007
UC-1-14"	Pad	1509.10	1498.98	1.007
A-10"	Nozzle	1203.20	1009.30	1.192
A-10"	Pad	1509.10	1498.98	1.007
MH-1-24"	Nozzle	1509.10	1498.98	1.007
MH-1-24"	Pad	1509.10	1498.98	1.007
C-14"	Nozzle	1509.10	1498.98	1.007
C-14"	Pad	1509.10	1498.98	1.007

Minimum 1.007

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H-1	1509.10	1498.98	1.007
S-1	1509.10	1498.98	1.007



S-2	1509.10	1498.98	1.007
S-3	1509.10	1498.98	1.007
S-4	1509.10	1498.98	1.007
S-5	1509.10	1498.98	1.007
S-6	1509.10	1498.98	1.007
S-7	1509.10	1498.98	1.007
H-2	1509.10	1498.98	1.007

Minimum

1.007

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
D-10"	818.37	2844.86	0.288
MH-3-24"	802.71	2844.86	0.282
E-8"	414.51	1881.28	0.220
UC-2-14"	776.66	2844.86	0.273
B-14"	776.66	2844.86	0.273
MH-2-24"	793.93	2844.86	0.279
UC-1-14"	768.13	2844.86	0.270
A-10"	408.60	1881.28	0.217
MH-1-24"	785.15	2844.86	0.276
C-14"	859.94	2844.86	0.302

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2023

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	6100	...	...	No Calc	No Calc
20	30	6100	...	...	No Calc	No Calc
30	40	6100	...	...	No Calc	No Calc
40	50	No Calc	2867	57.5	0.0027855	1002.4
50	60	20278.3	2870	59	0.00053048	523.646
60	70	20278.3	2870	59	0.00053048	523.646
70	80	20278.3	2870	59	0.00053048	523.646
80	90	20278.3	2870	59	0.00053048	523.646
90	100	20278.3	2870	59	0.00053048	523.646
100	110	20278.3	2870	59	0.00053048	523.646
110	120	20278.3	2870	59	0.00053048	523.646
120	130	No Calc	2866	57	0.0027623	1000.96

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kPa	External M.A.W.P. kPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	58.5	8.5438	103.421	2190.53
50	60	60	22.4342	103.421	1407.53
60	70	60	22.4342	103.421	1407.53
70	80	60	22.4342	103.421	1407.53
80	90	60	22.4342	103.421	1407.53
90	100	60	22.4342	103.421	1407.53
100	110	60	22.4342	103.421	1407.53
110	120	60	22.4342	103.421	1407.53
120	130	58	8.54117	103.421	2169.13

Minimum

1408

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	6100	No Calc	No Calc	No Calc
20	30	6100	No Calc	No Calc	No Calc
30	40	6100	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	20278.3	902456	No Calc	No Calc
60	70	20278.3	902456	No Calc	No Calc
70	80	20278.3	902456	No Calc	No Calc
80	90	20278.3	902456	No Calc	No Calc
90	100	20278.3	902456	No Calc	No Calc
100	110	20278.3	902456	No Calc	No Calc
110	120	20278.3	902456	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm³	Corroded Metal Wgt. kg	Corroded ID Volume cm³	Extra due Misc % kg
10	20	4358.87	...	4102.9	...	...
20	30	3455.34	...	3148.18	...	...
30	40	691.068	...	629.635	...	...
40	50	5242.32	3019284	5165.23	3025660	...
50	60	12315.5	17818726	12114.5	17844652	...
60	70	12315.5	17818726	12114.5	17844652	...
70	80	12315.5	17818726	12114.5	17844652	...
80	90	12315.5	17818726	12114.5	17844652	...
90	100	12315.5	17818726	12114.5	17844652	...
100	110	12315.5	17818726	12114.5	17844652	...
110	120	7060.87	10216071	6945.67	10230935	...
120	130	5242.32	3019284	5165.23	3025660	...
Total		99943	0.123E+09	97844	0.123E+09	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Plat	982.15	1473.19	2100	-1473.19	PL-6
10	Insl	2540.94	...	1250	...	FR-3
10	Lini	2412.77	...	1250	...	FR-3.1
10	Wght	1000	...	...	...	Taling Lug
20	Insl	3049.13	...	1500	...	FR-2
20	Lini	2895.32	...	1500	...	FR-2.1
20	Wght	500	...	0.00028956	...	Skirt MW
30	Insl	609.827	...	300	...	FR-1
30	Lini	579.064	...	300	...	FR-1.1
40	Liqd	2309.12	...	-343.75	...	Liquid 10
40	Insl	610.07	...	-318.75	...	Ins: 10
40	Lini	227.88	...	-318.75	...	clad
40	Nozl	466.917	...	-833.256	...	D-10"
40	Wght	246	2840	-685	...	PP_D
50	Plat	3931.29	0.21088E-04	0.48768E-04	...	PL-1
50	Liqd	13654.5	...	1500	...	Liquid 20
50	Insl	1398.02	...	1500	...	Ins: 20
50	Lini	623.457	...	1500	...	clad
50	Nozl	1453.43	-1625	1500	...	MH-3-24"
50	Nozl	389.114	1625	500	...	E-8"
50	Wght	1818	2815	500	...	PP_E
60	Plat	2039.64	...	1610	1516.12	PL-2
60	Pack	1063.14	...	2412.5	...	Bed#2
60	Liqd	682.724	...	75	...	Liquid: 30
60	Insl	1398.02	...	1500	...	Ins: 30
60	Lini	623.457	...	1500	...	clad
60	Nozl	479.063	1625	2200	...	UC-2-14"
60	Nozl	324.334	1625	600	...	B-14"
60	Wght	630	...	1825	...	DL_BS-1
60	Wght	3286	2891	600	...	PP_B
70	Plat	2039.64	...	2610	-1516.12	PL-3
70	Pack	2714.41	...	1500	...	Bed#2
70	Insl	1398.02	...	1500	...	Ins: 40
70	Lini	623.457	...	1500	...	clad
80	Pack	294.061	...	162.5	...	Bed#2
80	Pack	764.558	...	2577.5	...	Bed#1
80	Insl	1398.02	...	1500	...	Ins: 50
80	Lini	623.457	...	1500	...	clad
80	Nozl	1453.43	1625	1500	...	MH-2-24"
80	Wght	630	...	2155	...	DL_BS-2
80	Wght	800	...	799.999	...	DL_LD-2

80 Wght	160	...	325	...	DL_BL-2
80 Wght	1100	...	799.999	...	OP_LD-2
80 Wght	100	...	770	...	TA_LD-2
80 Wght	160	...	950	...	small Noz
80 Wght	4252	...	950	...	Cleat Weight
80 Wght	10000	...	950	...	Margin Weight
90 Plat	2039.64	...	610.001	1516.12	PL-4
90 Pack	2714.41	...	1500	...	Bed#1
90 Insl	1398.02	...	1500	...	Ins: 60
90 Lini	623.457	...	1500	...	clad
90 Nozl	479.063	1625	500	...	UC-1-14"
100 Plat	2039.64	...	1610	-1516.12	PL-5
100 Pack	592.646	...	327.5	...	Bed#1
100 Insl	1398.02	...	1500	...	Ins: 70
100 Lini	623.457	...	1500	...	clad
100 Nozl	311.064	1625	1850	...	A-10"
100 Wght	800	...	1110	...	DL_LD-1
100 Wght	160	...	655.001	...	DL_BL-1
100 Wght	250	...	1850	...	DL_FP
100 Wght	1100	...	1110	...	OP_LD-1
100 Wght	140	...	1850	...	OP_FP
100 Wght	100	...	1200	...	TA_LD-1
100 Wght	5182	2840	1850	...	PP_A
110 Insl	801.53	...	860	...	Ins: 80
110 Lini	357.449	...	860	...	clad
110 Nozl	1453.43	1625	860	...	MH-1-24"
110 Wght	230	...	1350	...	DL_PAD
110 Wght	200	...	1350	...	TA_PAD
110 Wght	600	1400	1720	...	Pipe Davit
120 Plat	8482.56	...	890.001	...	TOP PL
120 Insl	615.234	...	368.75	...	Ins: 90
120 Lini	227.88	...	368.75	...	clad
120 Nozl	377.955	...	958.5	...	C-14"
120 Wght	4218	2891	687	...	PP_C
120 Wght	2000	3500	642	...	PSV
120 Wght	1500	...	...	...	Lifting Lug
120 Forc	...	...	685	...	C_FxFZ
60 Pliq	1342.85	...	2412.500	...	
70 Pliq	3428.57	...	1500.000	...	
80 Pliq	371.43	...	162.500	...	
80 Pliq	965.71	...	2577.500	...	
90 Pliq	3428.57	...	1500.000	...	
100 Pliq	748.57	...	327.500	...	

Total Weight of Each Detail Type:

Platforms	21554.6
Packing	8143.2
Packing Liquid	10285.7
Liquid	16646.3
Insulation	16614.8
Lining	10441.1
Nozzles	7187.8
Weights	41162.0

Sum of the Detail Weights 132035.5 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	99943.7	99943.7	99943.7	99943.7	99943.7	99943.7
Nozzles	7187.8	7187.8	7187.8	7187.8	7187.8	7187.8
Wld Weights	18412.0	18412.0	18412.0	18412.0	18412.0	18412.0
Platforms	...	...	21554.6	21554.6	21554.6	21554.6
Packing	...	...	...	...	8143.2	8143.2
Insulation	...	...	16614.8	16614.8	16614.8	16614.8
Lining	10441.1	10441.1	10441.1	10441.1	10441.1	10441.1

*



Empty Weights	...	...	...	5660.0	5660.0	...
Test Weights	...	5660.0	...	...	...	...
Ope Weights	...	...	...	...	...	22750.0
Ope. Liquid	...	...	...	...	...	16646.3
Packing Liq	...	...	...	...	...	10285.7
Test Liquid	...	122544.6	...	...	...	...
Totals	135984.6	264189.2	174154.0	179814.0	187957.2	231979.2

Field Installation Options:

* Packing installed after lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	135984.6 kg
Shop Test Wt.	- Fabricated weight + water (full)	264189.2 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	174154.0 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	179814.0 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	187957.2 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	231979.2 kg
Field Test Wt.	- Empty weight + test medium (full)	302358.6 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	88010.1 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	123330.1 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	72879.7 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	207816	204989
20	30	249380	245987
30	40	49875.9	49197.4
40	50	94818.4	86296.1
50	60	270491	259181
60	70	270491	259181
70	80	270491	259181
80	90	270491	259181
90	100	270491	259181
100	110	270491	259181
110	120	155082	148597
120	130	94818.4	86296.1

Total	2474737.5	2376451.2 cm ²
Total	2663.8	2558.0 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	11294.7	11294.7	11294.7	1446.89	11294.7
20	30	9899.8	9899.8	9899.8	...	9899.8
30	40	1879.96	1879.96	1879.96	...	1879.96
40	50	6547.19	9102.31	9565.12	698.641	6793.19
50	60	20110.8	35583.3	37921.5	3388.16	21928.8
60	70	17810	24184.7	35620.7	10805.4	22159.1
70	80	16376.6	22519.6	34187.3	...	19091
80	90	31892.4	35388.1	49703.1	2361.83	34051
90	100	16855.7	22998.6	34666.4	778.478	19570.1
100	110	17997.7	25760.9	35808.4	15222.4	25012.3
110	120	10703.3	10703.3	20914.8	3201.83	10703.3
120	130	18446	22664	21463.9	19194.3	22664



Empty Support Force + the Sum of the Y forces 173060.38 kgf
Operating Support Force + the Sum of the Y forces 235511.31 kgf
Hydro Support Force + the Sum of the Y forces 306457.78 kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	205047	231979	302926
20	30	193752	220684	291631
30	40	183853	210785	281731
40	50	181973	208905	279851
50	60	175180	199802	270286
60	70	153251	164219	232365
70	80	131092	140034	196744
80	90	112001	117515	162557
90	100	77949.6	82126.7	112853
100	110	58379.5	59128.1	78187
110	120	33367.2	33367.2	42378.6
120	130	22664	22664	21463.9

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	57097.8	57097.8	57097.8
20	30	55650.9	55650.9	55650.9
30	40	55650.9	55650.9	55650.9
40	50	55650.9	55650.9	55650.9
50	60	54952.3	54952.3	54952.3
60	70	51564.1	51564.1	51564.1
70	80	40758.8	40758.8	40758.8
80	90	40758.8	40758.8	40758.8
90	100	38396.9	38396.9	38396.9
100	110	37618.5	37618.5	37618.5
110	120	22396.1	22396.1	22396.1
120	130	19194.3	19194.3	19194.3

Nozzle Flange MAWP Results: (kPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
D-10"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
MH-3-24"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
E-8"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
UC-2-14"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
B-14"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
MH-2-24"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
UC-1-14"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
A-10"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
MH-1-24"	6954.1	10340.0	430	600	GR 1.9	...	...	...	...
C-14"	10436.2	15510.0	430	900	GR 1.9	...	...	...	...
Min Rating	6954.1	10340.0, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2020 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.38731 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.35205 Hz.

The Natural Frequency for the Vessel (Filled) is 1.17796 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	3607.52	6355.57

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	3607.52	6355.57

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
120	685.00	2551.	-3532.	2551.	2747.		2747.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.186 \cdot 0.91) / (1 + 1.7 \cdot 3.4 \cdot 0.186)))$
 $= \min(0.85, 0.882)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	1.352 Hz
Natural Frequency of Vessel (Empty)	1.387 Hz
Natural Frequency of Vessel (Test)	1.178 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.610
Structure Height to Diameter ratio	8.830
Height to top of Structure	26805.500 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1250.0 + 300.0 = 1550.0$ mm
 $= 5.085$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (5.085 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15 / 274320.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 $= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 $= 16.0 \text{ psf } [78.119 \text{ kgf/m}^2]$

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 0.85 \cdot 0.61 \cdot 88.673$
 $= 735.8 \text{ lbs. } [333.8 \text{ kgf}]$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
K-3	1550.0	0.000	0.000	0.000	0.849	1.000	78.119
K-2	4300.0	0.000	0.000	0.000	0.849	1.000	78.119
K-1	6100.0	0.000	0.000	0.000	0.902	1.000	78.119
H-1	6936.9	0.000	0.000	0.000	0.927	1.000	78.119
S-1	7950.0	0.000	0.000	0.000	0.954	1.000	78.119
S-2	10950.0	0.000	0.000	0.000	1.020	1.000	78.119
S-3	13950.0	0.000	0.000	0.000	1.074	1.000	78.119
S-4	16950.0	0.000	0.000	0.000	1.119	1.000	78.119
S-5	19950.0	0.000	0.000	0.000	1.158	1.000	78.119
S-6	22950.0	0.000	0.000	0.000	1.192	1.000	78.119
S-7	25310.0	0.000	0.000	0.000	1.217	1.000	78.119
H-2	26568.9	0.000	0.000	0.000	1.230	1.000	78.119

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, m/sec
 Vc - Critical wind velocity, m/sec
 Vw - Maximum wind speed at top of structure, m/sec
 W - Total corroded weight of structure, kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/25920.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[10.02/85.04^2] \cdot 10^4 = 13.859$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.400E-06 no chance. [Vp]:
 $= W / (L \cdot Dr^2)$



$$= 229880 / (25920.0 \cdot 3056.651^2)$$

$$= 0.94923E-06$$

Since Vp is > 0.400E-06 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm	Pressure kgf/m ²	Force kgf	Cf
PL-6	15000.00	2400.00	78.12	117.18	1.00
PL-1	30000.00	6450.00	78.12	234.36	1.00
PL-2	30000.00	11060.00	78.12	234.36	1.00
PL-3	30000.00	15060.00	78.12	234.36	1.00
PL-4	30000.00	19060.00	78.12	234.36	1.00
PL-5	30000.00	23060.00	78.12	234.36	1.00
TOP PL	70000.00	27060.00	78.12	546.83	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure kgf/m ²	Force kgf
Taling Lug	0.00	300.00	78.12	0.00
Skirt MW	0.00	2800.00	78.12	0.00
PPD	0.00	5715.00	78.12	0.00
PPE	0.00	6950.00	78.12	0.00
DLBS-1	0.00	11275.00	78.12	0.00
PPB	0.00	10050.00	78.12	0.00
DLBS-2	0.00	17605.00	78.12	0.00
DLLD-2	0.00	16250.00	78.12	0.00
DLBL-2	0.00	15775.00	78.12	0.00
OPLD-2	0.00	16250.00	78.12	0.00
TALD-2	0.00	16220.00	78.12	0.00
small Noz	0.00	16400.00	78.12	0.00
Cleat Weight	0.00	16400.00	78.12	0.00
Margin Weight	0.00	16400.00	78.12	0.00
DLLD-1	0.00	22560.00	78.12	0.00
DLBL-1	0.00	22105.00	78.12	0.00
DLFP	0.00	23300.00	78.12	0.00
OPLD-1	0.00	22560.00	78.12	0.00
OPFP	0.00	23300.00	78.12	0.00
TALD-1	0.00	22650.00	78.12	0.00
PPA	0.00	23300.00	78.12	0.00
DLPAD	0.00	25800.00	78.12	0.00
TAPAD	0.00	25800.00	78.12	0.00
Pipe Davit	0.00	26170.00	78.12	0.00
PPC	0.00	26857.00	78.12	0.00
PSV	0.00	26812.00	78.12	0.00
Lifting Lug	0.00	26170.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 1.35205 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	1550	3295.2	82380	78.1189	566.727
20	30	4300	3295.2	98856	78.1189	440.575
30	40	6100	3295.2	19771.2	78.1189	88.115
40	50	6936.85	3776.4	28166.8	78.1189	125.532
50	60	7950	3780	113400	78.1189	714.546
60	70	10950	3780	113400	78.1189	765.776
70	80	13950	3780	113400	78.1189	783.698
80	90	16950	3780	113400	78.1189	505.394



90	100	19950	3780	113400	78.1189	751.685
100	110	22950	3780	113400	78.1189	764.421
110	120	25310	3780	65016	78.1189	289.759
120	130	26568.9	3775.2	28143	78.1189	738.058

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.000
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.3 * 0.147 = 0.191 \\ SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\ SDS &= 2/3 * Sms = 2/3 * 0.191 = 0.127 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(87.2950.75)$
 $= 0.571$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/1.352)$
 $= 0.740$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.571, 0.74)$ per 12.8.2
 $= 0.740$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.127 / (1.0 / 1.0)$
 $= 0.127$

Maximum value of Cs per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / Ie))$
 $= 0.065 / (0.74(1.0 / 1.0))$
 $= 0.088$

Minimum value of Cs per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot Ie, 0.030)$
 $= \max(0.044 \cdot 0.1274 \cdot 1.0, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= Cs \cdot W$ (Equation 12.8-1):
 $= 0.0879 \cdot 231979.2$
 $= 20387.111$ kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$



$$= 0.2 \cdot 0.127 \cdot 231979$$

$$= 5910.8 \text{ kgf}$$

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related
to the period of Vibration.

In this case, the value of k was 1.1198

The Natural Frequency for the Vessel (Ope...) is 1.35205 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	1250	11294.7	61.169	63.6569	2378.73
20	30	4000	9899.8	197.221	203.011	2090.94
30	40	5800	1879.96	56.7775	58.2406	1838.69
40	50	6125	9102.31	292.21	215.489	1790.79
50	60	7650	35583.3	1465.25	847.256	1558.86
60	70	10650	24184.7	1442.48	1083.43	652.204
70	80	13650	22519.6	1773.48	1312.34	35.9775
80	90	16650	35388.1	3481.31	3186.52	-537.821
90	100	19650	22998.6	2723.68	2024.26	-1439.51
100	110	22650	25760.9	3576.95	2530.79	-2025.52
110	120	25010	10703.3	1660.62	1680.18	-2681.9
120	130	25895	22664	3655.96	3009.6	-2954.62

The Earthquake Weight does not include the weight of the saddles.

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Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	3607.52	3607.52	99682.2	99682.2
20	30	4000	3607.52	3607.52	90663.4	90663.4
30	40	5800	3607.52	3607.52	79840.8	79840.8
40	50	6125	3607.52	3607.52	77676.3	77676.3
50	60	7650	3607.52	3607.52	77495.9	77495.9
60	70	10650	3607.52	3607.52	66673.3	66673.3
70	80	13650	3607.52	3607.52	55850.8	55850.8
80	90	16650	3607.52	3607.52	45028.2	45028.2
90	100	19650	3607.52	3607.52	34205.6	34205.6
100	110	22650	3607.52	3607.52	23383.1	23383.1
110	120	25010	3607.52	3607.52	12560.5	12560.5
120	130	25895	3607.52	3607.52	6355.57	6355.57

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	1250	6534.29	20387.1	95204.5	383837
20	30	4000	5967.56	20325.9	79577.2	332945
30	40	5800	5526.98	20128.7	62335.3	272263
40	50	6125	5438.87	20071.9	59045.6	260203
50	60	7650	5313.34	19779.7	58776.8	259207
60	70	10650	4598.79	18314.5	43908.6	202066
70	80	13650	3833.02	16872	31260.8	149286
80	90	16650	3049.32	15098.5	20937.3	101330
90	100	19650	2543.92	11617.2	12547.4	61256.3
100	110	22650	1792.24	8893.53	6043.18	30490.1
110	120	25010	1027.82	5316.59	1813.09	9174.89
120	130	25895	738.058	3655.96	294.437	1458.49

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	3607.52	0.030411	0.11978	0.94336E-04
20	30	3607.52	0.30092	0.55811	0.00019574
30	40	3607.52	0.61824	0.68116	0.00021435
40	50	3607.52	0.68652	0.69189	0.00021472
50	60	3607.52	1.02195	1.36723	0.000235
60	70	3607.52	1.72657	2.09884	0.00025223
70	80	3607.52	2.48289	2.87758	0.00026642
80	90	3607.52	3.28177	3.69432	0.00027757
90	100	3607.52	4.11408	4.53993	0.00028567
100	110	3607.52	4.9707	5.40527	0.00029072
110	120	3607.52	5.65569	5.90676	0.00029225
120	130	3607.52	5.91407	5.92138	0.00029225

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
Allowable deflection : 129.600 Actual deflection : 5.921 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	6534.29	0.02902	0.11277	0.87605E-04
20	30	5967.56	0.27879	0.50879	0.00017294
30	40	5526.98	0.56182	0.61712	0.00018801
40	50	5438.87	0.62182	0.62652	0.00018834
50	60	5313.34	0.91586	1.21763	0.00020493
60	70	4598.79	1.53009	1.85164	0.00021706
70	80	3833.02	2.1808	2.51626	0.00022546
80	90	3049.32	2.85681	3.2014	0.00023085
90	100	2543.92	3.54906	3.89891	0.0002338
100	110	1792.24	4.25025	4.60248	0.00023502
110	120	1027.82	4.80466	5.00691	0.0002352
120	130	738.058	5.01279	5.01867	0.0002352

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed m/sec	2nd Crit. Wind Speed m/sec
10	20	22.2165	138.853
20	30	22.2165	138.853
30	40	22.2165	138.853
40	50	25.4608	159.13
50	60	25.4851	159.282
60	70	25.4851	159.282
70	80	25.4851	159.282
80	90	25.4851	159.282
90	100	25.4851	159.282
100	110	25.4851	159.282
110	120	25.4851	159.282
120	130	25.4527	159.079



Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 129.600 Actual deflection : 5.019 mm

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 129.600 Actual deflection : 10.940 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	1486.1	1353.99	96976952	88409008
20	30	1486.1	1353.99	96976952	88409008
30	40	1486.1	1353.99	96976952	88409008
40	50	5161.55	5075.13	355164800	349461312
50	60	5296.73	5210.3	364482336	358784768
60	70	5296.73	5210.3	364482336	358784768
70	80	5296.73	5210.3	364482336	358784768
80	90	5296.73	5210.3	364482336	358784768
90	100	5296.73	5210.3	364482336	358784768
100	110	5296.73	5210.3	364482336	358784768
110	120	5296.73	5210.3	364482336	358784768
120	130	5116.53	5030.1	352061632	346356128



Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1181.99	2004.25	-1194.74	-1194.74
20	30	1181.99	2004.25	-1194.74	-1194.74
30	40	905.46	1706.92	-1068.8	-1375.58
40	50	1798.78	3413.83	-823.769	-1463.4
50	60	1798.78	3413.83	-827.029	-1465.25
60	70	1798.78	3413.83	-827.029	-1465.25
70	80	1798.78	3413.83	-827.029	-1465.25
80	90	1798.78	3413.83	-827.029	-1465.25
90	100	1798.78	3413.83	-827.029	-1465.25
100	110	1798.78	3413.83	-827.029	-1465.25
110	120	1798.78	3413.83	-827.029	-1465.25
120	130	1798.78	3413.83	-822.662	-1462.78

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	711.388	-13.4152	943.574
50	60	692.995	-13.0945	919.178
60	70	692.995	-13.0945	919.178
70	80	692.995	-13.0945	919.178
80	90	692.995	-13.0945	919.178
90	100	692.995	-13.0945	919.178
100	110	692.995	-13.0945	919.178
110	120	692.995	-13.0945	919.178
120	130	717.735	-13.5258	951.992

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-151.439	-171.33	-223.728	64.5837	64.5837
20	30	-143.097	-162.988	-215.386	62.9471	62.9471
30	40	-135.786	-155.677	-208.075	62.9471	62.9471
40	50	-35.8559	-37.8825	-35.8559	15.9248	15.9248
50	60	-33.6218	-35.5959	-33.6218	15.3162	15.3162
60	70	-29.4131	-31.3872	-29.4131	14.3719	14.3719
70	80	-25.1601	-26.8765	-25.1601	11.3602	11.3602
80	90	-21.496	-22.5544	-21.496	11.3602	11.3602
90	100	-14.9607	-15.7624	-14.9607	10.7019	10.7019
100	110	-11.2047	-11.3483	-11.2047	10.485	10.485
110	120	-6.4041	-6.4041	-6.4041	6.24221	6.24221
120	130	-4.50567	-4.50567	-4.50567	5.54177	5.54177

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	64.5837	107.686	434.16	53.8432	...
20	30	62.9471	90.0103	376.597	45.0051	...
30	40	62.9471	70.5079	307.959	35.254	...
40	50	15.9248	16.8962	74.4584	8.44808	...
50	60	15.3162	16.3822	72.2458	8.19109	...
60	70	14.3719	12.2381	56.3194	6.11907	...
70	80	11.3602	8.71298	41.6087	4.35649	...
80	90	11.3602	5.83562	28.2425	2.91781	...
90	100	10.7019	3.4972	17.0733	1.7486	...
100	110	10.485	1.68435	8.49816	0.84217	...
110	120	6.24221	0.50534	2.55721	0.25267	...
120	130	5.54177	0.08501	0.4211	0.042505	...



Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	350.324
20	30	...	...	...	304.563
30	40	...	...	...	250.101
40	50	...	...	...	60.539
50	60	...	...	...	58.7459
60	70	...	...	...	45.9939
70	80	...	...	...	34.049
80	90	...	...	...	23.1058
90	100	...	...	...	14.0435
100	110	...	...	...	7.15963
110	120	...	...	...	2.18017
120	130	...	...	...	0.34665

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	-2.60866	1.75683
20	30	-2.60866	1.54428
30	40	-2.60866	1.35798
40	50	-0.69596	0.35286
50	60	-0.67791	0.29919
60	70	-0.67791	0.12518
70	80	-0.67791	0.0069051
80	90	-0.67791	-0.10322
90	100	-0.67791	-0.27628
100	110	-0.67791	-0.38875
110	120	-0.67791	-0.51473
120	130	-0.70219	-0.58739

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	112.751	112.751	102.79	102.79
20	30	102.55	102.55	93.4896	93.4896
30	40	90.3085	90.3085	82.3297	82.3297
40	50	22.2274	22.2274	21.8705	21.8705
50	60	21.5996	21.5996	21.2619	21.2619
60	70	18.5831	18.5831	18.2926	18.2926
70	80	15.5667	15.5667	15.3233	15.3233
80	90	12.5502	12.5502	12.354	12.354
90	100	9.53375	9.53375	9.38472	9.38472
100	110	6.5173	6.5173	6.41542	6.41542
110	120	3.50085	3.50085	3.44612	3.44612
120	130	1.83498	1.83498	1.80524	1.80524

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...



40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
EP = Longitudinal Stress due to External Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads (Ope)
VE = Bending Stress due to Vortex Shedding Loads (Emp)
VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	130.97	1181.99	-439.07	1194.74	0.1108	0.3675
20	109.80	1181.99	-401.21	1194.74	0.0929	0.3358
30	85.37	905.46	-362.16	1068.80	0.0943	0.3388
40	18.50	1798.78	-91.60	823.77	0.0103	0.1112
50	19.00	1798.78	-87.60	827.03	0.0106	0.1059
60	15.10	1798.78	-75.28	827.03	0.0084	0.0910
70	9.80	1798.78	-61.48	827.03	0.0054	0.0743
80	7.57	1798.78	-51.92	827.03	0.0042	0.0628
90	8.09	1798.78	-39.37	827.03	0.0045	0.0476
100	6.80	1798.78	-30.57	827.03	0.0038	0.0370
110	3.17	1798.78	-17.33	827.03	0.0018	0.0210
120	2.25	1798.78	-12.67	822.66	0.0013	0.0154

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	377.98	1181.99	-677.34	1194.74	0.3198	0.5669
20	328.51	1181.99	-611.61	1194.74	0.2779	0.5119
30	268.93	905.46	-537.78	1068.80	0.2970	0.5032
40	63.19	1798.78	-134.19	823.77	0.0351	0.1629
50	62.34	1798.78	-128.98	827.03	0.0347	0.1560
60	49.66	1798.78	-108.24	827.03	0.0276	0.1309



70	35.82	1798.78	-86.13	827.03	0.0199	0.1041
80	25.42	1798.78	-68.62	827.03	0.0141	0.0830
90	19.04	1798.78	-49.52	827.03	0.0106	0.0599
100	12.57	1798.78	-35.76	827.03	0.0070	0.0432
110	5.00	1798.78	-18.84	827.03	0.0028	0.0228
120	2.63	1798.78	-12.82	822.66	0.0015	0.0156

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.08	1181.99	-458.96	1194.74	0.0940	0.3841
20	89.91	1181.99	-421.10	1194.74	0.0761	0.3525
30	65.48	905.46	-382.05	1068.80	0.0723	0.3575
40	16.47	1798.78	-93.63	823.77	0.0092	0.1137
50	17.02	1798.78	-89.57	827.03	0.0095	0.1083
60	13.13	1798.78	-77.26	827.03	0.0073	0.0934
70	8.09	1798.78	-63.19	827.03	0.0045	0.0764
80	6.51	1798.78	-52.98	827.03	0.0036	0.0641
90	7.29	1798.78	-40.17	827.03	0.0041	0.0486
100	6.66	1798.78	-30.71	827.03	0.0037	0.0371
110	3.17	1798.78	-17.33	827.03	0.0018	0.0210
120	2.25	1798.78	-12.67	822.66	0.0013	0.0154

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	441.92	1181.99	-781.07	1194.74	0.3739	0.6538
20	380.65	1181.99	-703.54	1194.74	0.3220	0.5889
30	306.90	905.46	-615.53	1068.80	0.3389	0.5759
40	75.08	1798.78	-150.14	823.77	0.0417	0.1823
50	73.86	1798.78	-144.46	827.03	0.0411	0.1747
60	58.01	1798.78	-120.54	827.03	0.0323	0.1457
70	41.67	1798.78	-95.41	827.03	0.0232	0.1154
80	29.50	1798.78	-74.81	827.03	0.0164	0.0905
90	21.27	1798.78	-53.35	827.03	0.0118	0.0645
100	13.76	1798.78	-37.24	827.03	0.0077	0.0450
110	5.38	1798.78	-19.22	827.03	0.0030	0.0232
120	2.70	1798.78	-12.89	822.66	0.0015	0.0157

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-342.15	1194.74		0.2864
20		2004.25	-323.34	1194.74		0.2706
30		1706.92	-306.28	1375.58		0.2227
40		3413.83	-60.23	1463.40		0.0412
50		3413.83	-57.13	1465.25		0.0390
60		3413.83	-49.90	1465.25		0.0341
70		3413.83	-40.88	1465.25		0.0279
80		3413.83	-35.77	1465.25		0.0244
90		3413.83	-27.41	1465.25		0.0187
100	0.12	3413.83	-22.53	1465.25	0.0000	0.0154
110	0.09	3413.83	-12.90	1465.25	0.0000	0.0088
120	1.08	3413.83	-10.09	1462.78	0.0003	0.0069

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-288.31	1194.74		0.2413
20		2004.25	-278.33	1194.74		0.2330
30		1706.92	-271.02	1375.58		0.1970
40		3413.83	-51.78	1463.40		0.0354
50		3413.83	-48.94	1465.25		0.0334
60		3413.83	-43.78	1465.25		0.0299
70		3413.83	-36.52	1465.25		0.0249



80		3413.83	-32.86	1465.25		0.0224
90		3413.83	-25.66	1465.25		0.0175
100		3413.83	-21.69	1465.25		0.0148
110		3413.83	-12.65	1465.25		0.0086
120	1.04	3413.83	-10.05	1462.78	0.0003	0.0069

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.08	1181.99	-458.96	1194.74	0.0940	0.3841
20	89.91	1181.99	-421.10	1194.74	0.0761	0.3525
30	65.48	905.46	-382.05	1068.80	0.0723	0.3575
40	727.86	1798.78		823.77	0.4046	
50	710.02	1798.78		827.03	0.3947	
60	706.12	1798.78		827.03	0.3926	
70	701.08	1798.78		827.03	0.3898	
80	699.51	1798.78		827.03	0.3889	
90	700.29	1798.78		827.03	0.3893	
100	699.66	1798.78		827.03	0.3890	
110	696.16	1798.78		827.03	0.3870	
120	719.99	1798.78		822.66	0.4003	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	441.92	1181.99	-781.07	1194.74	0.3739	0.6538
20	380.65	1181.99	-703.54	1194.74	0.3220	0.5889
30	306.90	905.46	-615.53	1068.80	0.3389	0.5759
40	786.47	1798.78		823.77	0.4372	
50	766.86	1798.78		827.03	0.4263	
60	751.01	1798.78		827.03	0.4175	
70	734.66	1798.78		827.03	0.4084	
80	722.49	1798.78		827.03	0.4017	
90	714.27	1798.78		827.03	0.3971	
100	706.76	1798.78		827.03	0.3929	
110	698.38	1798.78		827.03	0.3883	
120	720.44	1798.78		822.66	0.4005	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	111.08	1181.99	-458.96	1194.74	0.0940	0.3841
20	89.91	1181.99	-421.10	1194.74	0.0761	0.3525
30	65.48	905.46	-382.05	1068.80	0.0723	0.3575
40	3.05	1798.78	-107.04	1309.51	0.0017	0.0817
50	3.93	1798.78	-102.67	1313.38	0.0022	0.0782
60	0.03	1798.78	-90.35	1313.38	0.0000	0.0688
70		1798.78	-76.29	1313.38		0.0581
80		1798.78	-66.07	1313.38		0.0503
90		1798.78	-53.27	1313.38		0.0406
100		1798.78	-43.81	1313.38		0.0334
110		1798.78	-30.42	1313.38		0.0232
120		1798.78	-26.20	1308.19		0.0200

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	441.92	1181.99	-781.07	1194.74	0.3739	0.6538
20	380.65	1181.99	-703.54	1194.74	0.3220	0.5889
30	306.90	905.46	-615.53	1068.80	0.3389	0.5759
40	61.67	1798.78	-163.56	1309.51	0.0343	0.1249
50	60.77	1798.78	-157.55	1313.38	0.0338	0.1200
60	44.92	1798.78	-133.63	1313.38	0.0250	0.1017
70	28.57	1798.78	-108.50	1313.38	0.0159	0.0826
80	16.40	1798.78	-87.91	1313.38	0.0091	0.0669



90	8.18	1798.78	-66.44	1313.38	0.0045	0.0506
100	0.67	1798.78	-50.33	1313.38	0.0004	0.0383
110		1798.78	-32.31	1313.38		0.0246
120		1798.78	-26.42	1308.19		0.0202

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-342.15	1194.74		0.2864
20		2004.25	-323.34	1194.74		0.2706
30		1706.92	-306.28	1375.58		0.2227
40	932.09	3413.83		1463.40	0.2730	
50	909.06	3413.83		1465.25	0.2663	
60	910.26	3413.83		1465.25	0.2666	
70	909.73	3413.83		1465.25	0.2665	
80	911.96	3413.83		1465.25	0.2671	
90	916.67	3413.83		1465.25	0.2685	
100	919.30	3413.83		1465.25	0.2693	
110	919.27	3413.83		1465.25	0.2693	
120	953.07	3413.83		1462.78	0.2792	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		2004.25	-288.31	1194.74		0.2413
20		2004.25	-278.33	1194.74		0.2330
30		1706.92	-271.02	1375.58		0.1970
40	923.64	3413.83		1463.40	0.2706	
50	900.87	3413.83		1465.25	0.2639	
60	904.14	3413.83		1465.25	0.2648	
70	905.38	3413.83		1465.25	0.2652	
80	909.04	3413.83		1465.25	0.2663	
90	914.92	3413.83		1465.25	0.2680	
100	918.46	3413.83		1465.25	0.2690	
110	919.02	3413.83		1465.25	0.2692	
120	953.03	3413.83		1462.78	0.2792	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-216.02	1194.74		0.1808
20		1181.99	-206.04	1194.74		0.1725
30		905.46	-198.73	1068.80		0.1859
40	691.46	1798.78		823.77	0.3844	
50	674.69	1798.78		827.03	0.3751	
60	677.95	1798.78		827.03	0.3769	
70	679.20	1798.78		827.03	0.3776	
80	682.86	1798.78		827.03	0.3796	
90	688.74	1798.78		827.03	0.3829	
100	692.28	1798.78		827.03	0.3849	
110	692.83	1798.78		827.03	0.3852	
120	718.77	1798.78		822.66	0.3996	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-156.10	1194.74		0.1307
20		1181.99	-148.50	1194.74		0.1243
30		905.46	-141.84	1068.80		0.1327
40	674.14	1798.78		823.77	0.3748	
50	657.98	1798.78		827.03	0.3658	
60	662.12	1798.78		827.03	0.3681	
70	666.56	1798.78		827.03	0.3706	
80	670.81	1798.78		827.03	0.3729	
90	677.49	1798.78		827.03	0.3766	



100	681.83	1798.78	827.03	0.3791
110	686.70	1798.78	827.03	0.3818
120	713.31	1798.78	822.66	0.3966

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-235.91	1194.74		0.1975
20		1181.99	-225.94	1194.74		0.1891
30		905.46	-218.62	1068.80		0.2046
40	689.43	1798.78		823.77	0.3833	
50	672.72	1798.78		827.03	0.3740	
60	675.98	1798.78		827.03	0.3758	
70	677.48	1798.78		827.03	0.3766	
80	681.80	1798.78		827.03	0.3790	
90	687.93	1798.78		827.03	0.3824	
100	692.13	1798.78		827.03	0.3848	
110	692.83	1798.78		827.03	0.3852	
120	718.77	1798.78		822.66	0.3996	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-216.02	1194.74		0.1808
20		1181.99	-206.04	1194.74		0.1725
30		905.46	-198.73	1068.80		0.1859
40	691.46	1798.78		823.77	0.3844	
50	674.69	1798.78		827.03	0.3751	
60	677.95	1798.78		827.03	0.3769	
70	679.20	1798.78		827.03	0.3776	
80	682.86	1798.78		827.03	0.3796	
90	688.74	1798.78		827.03	0.3829	
100	692.28	1798.78		827.03	0.3849	
110	692.83	1798.78		827.03	0.3852	
120	718.77	1798.78		822.66	0.3996	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-235.91	1194.74		0.1975
20		1181.99	-225.94	1194.74		0.1891
30		905.46	-218.62	1068.80		0.2046
40		1798.78	-53.81	823.77		0.0653
50		1798.78	-50.91	827.03		0.0616
60		1798.78	-45.76	827.03		0.0553
70		1798.78	-38.24	827.03		0.0462
80		1798.78	-33.91	827.03		0.0410
90		1798.78	-26.46	827.03		0.0320
100		1798.78	-21.83	827.03		0.0264
110		1798.78	-12.65	827.03		0.0153
120	1.04	1798.78	-10.05	822.66	0.0006	0.0122

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	7.76	984.99	-346.91	995.62	0.0079	0.3484
20	4.05	984.99	-326.94	995.62	0.0041	0.3284
30		754.55	-307.57	890.67		0.3453
40	712.01	1498.98		686.47	0.4750	
50	694.61	1498.98		689.19	0.4634	
60	694.69	1498.98		689.19	0.4634	
70	693.05	1498.98		689.19	0.4623	
80	694.25	1498.98		689.19	0.4631	
90	697.19	1498.98		689.19	0.4651	
100	698.26	1498.98		689.19	0.4658	



110	695.82	1498.98	689.19	0.4642
120	720.02	1498.98	685.55	0.4803

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	7.76	984.99	-346.91	995.62	0.0079	0.3484
20	4.05	984.99	-326.94	995.62	0.0041	0.3284
30		754.55	-307.57	890.67		0.3453
40		1498.98	-89.10	1091.26		0.0816
50		1498.98	-85.31	1094.49		0.0779
60		1498.98	-77.31	1094.49		0.0706
70		1498.98	-66.89	1094.49		0.0611
80		1498.98	-59.66	1094.49		0.0545
90		1498.98	-49.37	1094.49		0.0451
100		1498.98	-41.83	1094.49		0.0382
110		1498.98	-29.76	1094.49		0.0272
120		1498.98	-26.00	1090.16		0.0238

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		1181.99	-235.91	1194.74		0.1975
20		1181.99	-225.94	1194.74		0.1891
30		905.46	-218.62	1068.80		0.2046
40	689.43	1798.78		823.77	0.3833	
50	672.72	1798.78		827.03	0.3740	
60	675.98	1798.78		827.03	0.3758	
70	677.48	1798.78		827.03	0.3766	
80	681.80	1798.78		827.03	0.3790	
90	687.93	1798.78		827.03	0.3824	
100	692.13	1798.78		827.03	0.3848	
110	692.83	1798.78		827.03	0.3852	
120	718.77	1798.78		822.66	0.3996	

Absolute Maximum of the all of the Stress Ratio's 0.6538

Governing Element: K-3

Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
K-3	...	-169.6	64.6	546.9	441.9	+ Tens
K-3	...	-169.6	-64.6	-546.9	-781.1	- Comp
K-2	...	-161.4	62.9	479.1	380.7	+ Tens
K-2	...	-161.4	-62.9	-479.1	-703.5	- Comp
K-1	...	-154.3	62.9	398.3	306.9	+ Tens
K-1	...	-154.3	-62.9	-398.3	-615.5	- Comp
H-1	...	-37.5	15.9	96.7	75.1	+ Tens
H-1	...	-37.5	-15.9	-96.7	-150.1	- Comp
S-1	...	-35.3	15.3	93.8	73.9	+ Tens
S-1	...	-35.3	-15.3	-93.8	-144.5	- Comp
S-2	...	-31.3	14.4	74.9	58.0	+ Tens
S-2	...	-31.3	-14.4	-74.9	-120.5	- Comp
S-3	...	-26.9	11.4	57.2	41.7	+ Tens
S-3	...	-26.9	-11.4	-57.2	-95.4	- Comp
S-4	...	-22.7	11.4	40.8	29.5	+ Tens
S-4	...	-22.7	-11.4	-40.8	-74.8	- Comp



S-5	...	-16.0	10.7	26.6	21.3	+ Tens
S-5	...	-16.0	-10.7	-26.6	-53.3	- Comp
S-6	...	-11.7	10.5	15.0	13.8	+ Tens
S-6	...	-11.7	-10.5	-15.0	-37.2	- Comp
S-7	...	-6.9	6.2	6.1	5.4	+ Tens
S-7	...	-6.9	-6.2	-6.1	-19.2	- Comp
H-2	...	-5.1	5.5	2.3	2.7	+ Tens
H-2	...	-5.1	-5.5	-2.3	-12.9	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: K-1

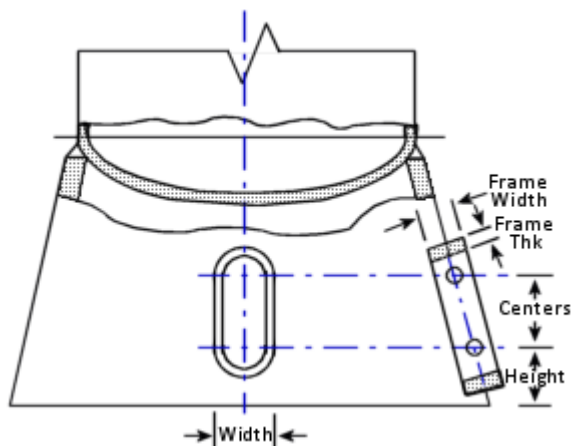
Total vessel (empty) weight	Fg	206827.641 kgf
Head Weight below section 2-2	DeltaFg	6793.189 kgf
Weight of vessel contents (liquid)	Ff	16646.326 kgf
Vessel Gross Weight	Fg + Ff	223473.969 kgf
Skirt Mean Diameter	Dz	2628.000 mm
Skirt Analysis Thickness (Corroded)	ez	16.400 mm
Skirt Height	H	600.000 mm
Additional Axial Force at section 1-1	F1	-3532.100 kgf
Moment at section 1-1	M1	337879.500 kgf-m.
Applied Moment - Bottom of Skirt	Mb	352104.250 kgf-m.
Weight at Bottom of Skirt	W	210784.688 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2809.500 mm
Head Corroded Thickness	eb	57.500 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 688.5 \text{ mm} \\
 K &= D_i / (2 * h_i) = 2752.0 / (2 * 688.5) = 1.9985 \\
 R &= D_i (0.44 * K + 0.02) = 2752.0 * (0.44 * 1.9985 + 0.02) = 2475.0417 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 2752.0 * ((0.5 / 1.9985) - 0.08) = 468.3401 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	11689059.000 cm ⁴
Moment of Inertia	Iyy =	11689059.000 cm ⁴
Section Modulus	W =	88957904.000 mm ³
Section Metal Area	A =	1354.001 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(11689059, 11689059) / 1354.001)} \\
 &= 929.138 \text{ mm}
 \end{aligned}$$



Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 929.138/600.0$$

$$= 1.549$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 337879.5 kgf-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -337879.5 / 35583E+09 - 210784.688 / 1354.00146$$

$$= -250.63 \text{ kgf/cm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +337879.5 / 35583E+09 - 210784.688 / 1354.00146$$

$$= -60.721 \text{ kgf/cm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= -3532.1 - 206827.641 - 16646.326 + 4 \cdot 337879.5 / 2628.0$$

$$= 287269.6 \text{ kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= -3532.1 - 206827.641 - 16646.326 - 4 \cdot 337879.5 / 2628.0$$

$$= -741281.7 \text{ kgf}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (287269.562 + 6793.189 + 16646.326) / (\pi \cdot 2809.5 \cdot 57.5) + 5929.49 \cdot 2809.5 / (4 \cdot 57.5)$$

$$= 799.82 \text{ kgf/cm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-0.7E+06 + 6793.189 + 16646.326) / (\pi \cdot 2809.5 \cdot 57.5) + 5929.49 \cdot 2809.5 / (4 \cdot 57.5)$$

$$= 597.15 \text{ kgf/cm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (16646.326 + 6793.189) / (\pi \cdot 2809.5 \cdot 57.5) + 5929.49 \cdot 2809.5 / (4 \cdot 57.5)$$

$$= 743.22 \text{ kgf/cm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 287269.562 / (\pi \cdot 2628.0 \cdot 16.4)$$

$$= 212.16 \text{ kgf/cm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -741281.688 / (\pi \cdot 2628.0 \cdot 16.4)$$

$$= -547.48 \text{ kgf/cm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(2809.5 + 57.5 - 2628.0 + 16.4) / (468.34 + 57.5) \\ = 0.7572$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.757) = 40.786 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ = 0.5 \cdot \sqrt{(57.5^2 + 16.4^2 + 2 \cdot 57.5 \cdot 16.4 \cdot 0.757)} \\ = 35.367 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 35.36 \cdot 287269.6 = 10159.77 \text{ kgf-m.} \\ M_q = a \cdot F_{zq} = 35.36 \cdot -741281.7 = -26216.67 \text{ kgf-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2 \\ = 0.63 - 0.057(57.5/16.4)^2 \\ = -0.071$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ = -0.071 \cdot 6 \cdot 10159.771 / (\pi \cdot 2809.5 \cdot 57.5^2) \\ = -14.77 \text{ kgf/cm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ = -0.071 \cdot 6 \cdot -26216.67 / (\pi \cdot 2809.5 \cdot 57.5^2) \\ = 38.10 \text{ kgf/cm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ = -0.071 \cdot 6 \cdot 10159.771 / (\pi \cdot 2628.0 \cdot 16.4^2) \\ = -194.04 \text{ kgf/cm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ = -0.071 \cdot 6 \cdot -26216.67 / (\pi \cdot 2628.0 \cdot 16.4^2) \\ = 500.72 \text{ kgf/cm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 2.264/110.61 = 0.0205 \\ y = 125 \cdot eb/D_b = 125 \cdot 57.5/2809.5 = 2.5583$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 2.558 = 3.6883$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b (y / y_A) \cdot \alpha - 1 / (4 \cdot eb) \\ = (5929.49 + 0.0) \cdot 2809.5 (40.79/40.0) \cdot 3.688 - 1 / (4 \cdot 57.5) \\ = 2039.16 \text{ kgf/cm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ = 799.82 - -14.765 + 2039.16 \\ = 2853.75 \text{ kgf/cm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ = 799.82 + -14.765 - 2039.16 \\ = -1254.11 \text{ kgf/cm}^2$$



Stress [σ_{1qitot}]:
= $\sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$
= $597.154 - 38.101 + 2039.16$
= 2598.21 kgf/cm^2

Stress [σ_{1qotot}]:
= $\sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$
= $597.154 + 38.101 - 2039.16$
= $-1403.90 \text{ kgf/cm}^2$

Stress [σ_{2pitot}]:
= $\sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$
= $743.216 + -14.765 + 2039.16$
= 2767.61 kgf/cm^2

Stress [σ_{2potot}]:
= $\sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
= $743.216 - -14.765 - 2039.16$
= $-1281.18 \text{ kgf/cm}^2$

Stress [σ_{2qitot}]:
= $\sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$
= $743.216 + 38.101 + 2039.16$
= 2820.48 kgf/cm^2

Stress [σ_{2qotot}]:
= $\sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
= $743.216 - 38.101 - 2039.16$
= $-1334.04 \text{ kgf/cm}^2$

Stress [σ_{3pitot}]:
= $\sigma_{3pm} - \sigma_{3pb}$
= $212.164 - -194.044$
= 406.21 kgf/cm^2

Stress [σ_{3potot}]:
= $\sigma_{3pm} + \sigma_{3pb}$
= $212.164 + -194.044$
= 18.12 kgf/cm^2

Stress [σ_{3qitot}]:
= $\sigma_{3qm} - \sigma_{3qb}$
= $-547.476 - 500.718$
= $-1048.19 \text{ kgf/cm}^2$

Stress [σ_{3qotot}]:
= $\sigma_{3qm} + \sigma_{3qb}$
= $-547.476 + 500.718$
= -46.76 kgf/cm^2

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
= $fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$
= $1498.98(3 - (1 / 1.5)) \cdot (799.82 / 1509.1)^2$
= 4216.23 kgf/cm^2

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
= $fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$
= $1498.98(3 - (1 / 1.5)) \cdot (597.154 / 1509.1)^2$
= 4340.47 kgf/cm^2

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
= $fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$
= $1498.98(3 - (1 / 1.5)) \cdot (743.216 / 1509.1)^2$
= 4254.56 kgf/cm^2

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:
= $fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$



$$= 1498.98(3 - (1/1.5) \cdot (743.216/1509.1)^2)$$

$$= 4254.56 \text{ kgf/cm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1/1.5) \cdot (\sigma_{3pm}/f)^2)$$

$$= 1498.98(3 - (1/1.5) \cdot (212.164/1509.1)^2)$$

$$= 4477.19 \text{ kgf/cm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1/1.5) \cdot (\sigma_{3qm}/f)^2)$$

$$= 1498.98(3 - (1/1.5) \cdot (-547.476/1509.1)^2)$$

$$= 4365.42 \text{ kgf/cm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	2853.75	4216.23	Pass	46
Stot1po	-1254.11	4216.23	Pass	47
Stot1qi	2598.21	4340.47	Pass	48
Stot1qo	-1403.90	4340.47	Pass	49
Stot2pi	2767.61	4254.56	Pass	50
Stot2po	-1281.18	4254.56	Pass	51
Stot2qi	2820.48	4254.56	Pass	52
Stot2qo	-1334.04	4254.56	Pass	53
Stot3pi	406.21	4477.19	Pass	54
Stot3po	18.12	4477.19	Pass	55
Stot3qi	-1048.19	4365.42	Pass	56
Stot3qo	-46.76	4365.42	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
 Packing is installed in the Field after being lifted.
 Insulation is installed in the Shop.
 Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	18092.260 mm
Center of Gravity of the Packing	16390.000 mm
Center of Gravity of the Liquid	7451.902 mm
Center of Gravity of the Insulation	11177.094 mm
Center of Gravity of the Lining	8702.585 mm
Center of Gravity of the Nozzles	15518.511 mm
Center of Gravity of the Added Weights (Operating)	17915.561 mm
Center of Gravity of the Added Weights (Empty)	17261.775 mm
Center of Gravity of Bare Shell New and Cold	14880.459 mm
Center of Gravity of Bare Shell Corroded	14938.947 mm
Vessel CG in the Operating Condition	14730.493 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	14908.841 mm
Vessel CG in the Test Condition	15356.364 mm

Warning: Lifting Lug is Higher than top of vessel!

Rigging Analysis Results:

- Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	25920.00 mm
Total vessel weight (No Liquid) Twt	187957.23 kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	281935.84 kgf
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	36300.00 mm
Design Reaction force at the tailing lug	166141.44 kgf
Design Reaction force at the lifting lug	115794.40 kgf
CG distance during field lifting operation	14908.84 mm
CG distance from tailing lug	14908.84 mm
CG distance from the nearer lifting lug/trunnion	14908.84 mm
Compressive Allowable Increase Factor	1.00

Critical Values: (kgf/cm², mm)

	Max Stress	Elevation	Allowables
Bending	868.75	5980.00	1152.08 (UG-23)
Shear	93.69	0.00	984.99 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment kgf-m.	Bending Stress kgf/cm ²	Shear Force kgf	Shear Stress kgf/cm ²
0.00	0.7	0.0	139225.3	93.7
5620.00	797335.0	822.2	105404.2	70.9
7350.00	993240.7	272.5	89064.4	16.8
13950.00	1469014.0	403.0	26813.8	5.1
20550.01	1465773.1	402.2	54605.0	10.3
25870.01	1125502.5	319.7	104891.8	20.5



Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.7541 at elevation 5980.0 mm - Must be ≤ 1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2646.0000	mm
Skirt Thickness	STHK	18.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	44.0000	mm
Design Temperature of the Basing		82.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	1407.13	kgf/cm²
Basing Yield Stress		2473.72	kgf/cm²
Inside Diameter of Basing	DI, D4	2462.0000	mm
Outside Diameter of Basing	DOU, D5	2982.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	1407.21	kgf/cm²
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	2814.0000	mm
Bolt Allowable Shear Stress		844.324	kgf/cm²

Ultimate Comp. Strength of Concrete	FPC	421.8	kgf/cm²
Allowable Comp. Strength of Concrete	FC	229.4	kgf/cm²
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	158.0000	mm
Width of Gussets at Base Plate	BWDT	158.0000	mm
Gusset Plate Elastic Modulus	E	2031470.0	kgf/cm²
Gusset Plate Yield Stress	SY	2473.7	kgf/cm²
Height of Gussets	HG, h1	274.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	168.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm

External Corrosion Allowance	CA	0.0000	mm
------------------------------	----	--------	----

Dead Weight of Vessel	DW	191489.3	kgf
Operating Weight of Vessel	ROW	235511.3	kgf
Earthquake Moment on Basing	EQMOM	433904.2	kgf-m.
Wind Moment on Basing	WIMOM	145271.8	kgf-m.

User Moment at the Base [Earthquake Case]		99682.2	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		99682.2	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
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Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 533586/2814.0) - 229600)/20$$

$$= 26443.6 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 244954/2814.0) - 191489)/20$$

$$= 7835.2 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/2814.0) - 235511)/20$$

$$= -11775.6 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 11.5517 cm²
Area Available in all the Bolts Abss * RN : 231.0331 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.451	1.511	2.441	0.438	0.781	114129.0	343729.5	8203.2
4	0.376	0.338	1.705	2.277	0.421	0.784	118580.7	348181.2	9135.7
6	0.319	0.310	1.561	2.400	0.433	0.782	115214.9	344815.4	8422.2
8	0.305	0.303	1.523	2.431	0.437	0.781	114398.8	343999.3	8257.2
10	0.304	0.305	1.520	2.433	0.437	0.781	114331.2	343931.7	8243.6
12	0.304	0.304	1.521	2.433	0.437	0.781	114348.1	343948.6	8247.0
13	0.304	0.304	1.520	2.433	0.437	0.781	114339.7	343940.2	8245.3

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 114339.7 / (2.613 \cdot 2814.0 \cdot 2.433)$$

$$= 1278.029 \text{ kgf/cm}^2, \text{ Should be } \leq 1407.2 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 11.552 / (\pi \cdot 2814.0)$$

$$= 2.613 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 56.796[(2 \cdot 0.3041 \cdot 2814.0 + 260.0) / (2 \cdot 0.3041 \cdot 2814.0)]$$

$$= 65.423 \text{ kgf/cm}^2, \text{ Should be } \leq 229.4 \text{ kgf/cm}^2$$

Values for table 10.3, l = 168.0, b = 221.011, l/b = 0.760143

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(2154.205, 3632.305) = 3632.305 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 3632.305 / 1632.7)} + 0.0$$

$$= 36.536 \text{ mm}$$

Nomenclature:

a = (Dc-Ds)/2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 168.0, b = 96.0, b/l = 0.571429

As b/l (0.571) is less than 1, inverting b/l = 1.75.

Moment Term, based on geometry [Mo]:

$$= P/(4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)/(\pi \cdot g))) + 1)] - [(0.7-g2)P/(4\pi)]$$

$$= 16255.58/(4 \cdot \pi) [1.3(\ln((2 \cdot 168.0 \cdot \sin(\pi \cdot 84.0/168.0))$$

$$(\pi \cdot 40.0)) + 1)] - [(0.7 - 0.044) \cdot 16255.58/(4 \cdot \pi)]$$



= 2099.5261 kgf

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(2099.53) / 1649.15) + 0.0}$$

$$= 27.6380 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 16255.58 / (1265.526 \cdot 158.0) + 0.0$$

$$= 9.525 \text{ (not less than } 9.525 + 0.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 2814.0 / 20 = 442.022 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (16256 \cdot 84.0 / (442.02 \cdot 318.0 \cdot 2111))^{2/3} \cdot 1323.0^{1/3} + \text{Ca}$$

$$= 11.517 + 1.6 = 13.117 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 23994.6 / (11.552 \cdot 20)$$

$$= 103.9 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	36.5359	mm
Actual Basing Thickness as entered by user	44.0000	mm
Required Thickness of Chair Cap	27.6380	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	9.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	13.1174	mm
Given Thickness of Skirt	18.0000	mm
Required Gusset Height to meet local stress	118.7615	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 781.068 \cdot (18.0 - 1.6)$$

$$= 128.1 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 128 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707]$$

$$= 9.155 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 16255.6 / (172.0 + 2 \cdot (274.0 + 44.0))$$

$$= 20.118 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 16255.6 \cdot 84.0 / (172.0 \cdot (318.0) + 0.6667 \cdot (318.0)^2)$$

$$= 11.182 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{20.12^2 + 11.18^2}$$

$$= 23.017 \text{ kgf/mm}$$



Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 23.02 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707] \\ &= 1.645 \text{ mm} \end{aligned}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 16255.6 / (2 \cdot 158.0) \\ &= 51.442 \text{ kgf/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 16255.6 \cdot 84.0 / (2 \cdot 318.0 \cdot 158.0) \\ &= 13.588 \text{ kgf/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{51.44^2 + 13.59^2} \\ &= 53.206 \text{ kgf/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 53.21 / [(0.4 \cdot 2474) \cdot 2 \cdot 0.707] \\ &= 3.803 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	9.1553	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.8028	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D-10"

From: 40

Pressure for Reinforcement Calculations	P	5958.832	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Elliptical Head	D	2750.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	58.5000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material [Impact Tested]		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	55.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1875.0	mm
Centerline Distance to Tangent		1650.0	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: D-10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750	in.
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Actual Thickness Used in Calculation 0.520 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 5958.83 \cdot 2752.0 \cdot 0.8994 / (2 \cdot 1498.98 \cdot 1.0 - 0.2 \cdot 5958.83)$
 $= 50.3708 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5958.83 \cdot 136.525 / (1499 \cdot 1.0 + 0.4 \cdot 5958.83)$
 $= 5.4460 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8541 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 66.798 / 0.5773$
 $= 115.6982 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	493.2934	mm
Parallel to Vessel Wall, opening length	d	246.6467	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	143.7500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499.0/1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499.0/1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	124.238	9.303	NA
Area in Shell	A1	17.584	123.215	NA
Area in Nozzle Wall	A2	22.297	35.499	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	168.799	168.799	NA
TOTAL AREA AVAILABLE	Atot	209.680	328.513	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (246.6467 \cdot 50.3708 \cdot 1.0 + 2 \cdot 80.0 \cdot 50.3708 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 124.238 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 246.647 (1.0 \cdot 57.5 - 1.0 \cdot 50.371) - 2 \cdot 80.0$$

$$(1.0 \cdot 57.5 - 1.0 \cdot 50.3708) \cdot (1 - 1.0)$$

$$= 17.584 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 143.75) (13.2 - 5.45) 1.0$$

$$= 22.297 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(143.8, 250.0, 100.0)) \cdot (80.0 - 13.2) \cdot 1.0$$

$$= 168.799 \text{ cm}^2 \text{ Includes Bevel Area of } 35.202 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.4460 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 56.9522 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 56.9522 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9672 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.103, \max(56.9522, 2.5)]$$

$$= 8.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.446, 8.1026)$$

$$= 8.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 560.5,	Allowable	: 1499.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3199.6 kgf/cm ²	Passed
Occasional	: 269.4,	Allowable	: 1993.6 kgf/cm ²	Passed
Shear	: 162.0,	Allowable	: 1049.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 13.202, tr = 5.41, c = 0.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.41, Temp. Reduction = 48 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -19 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 5958.83/10340.00 = 0.576

Weld Size Calculations, Description: D-10"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (124.2378 - 17.5841 + 2 \cdot 80.0 \cdot 1.0 \cdot (1.0 \cdot 57.5 - 50.3708)) \cdot 1499)$
 $= 176970.12 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (191.096 + 1. - 0.0 \cdot 1.0) \cdot 1499$
 $= 287947.69 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (191.096 + 0.0 + 1. + (92.0)) \cdot 1499$
 $= 425853.66 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (191.096 + 0.0 + 1. + 0.0 + (92.0)) \cdot 1499$
 $= 425853.66 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 1499$
 $= 31503. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 129.9242) \cdot (80.0 - 0.0) \cdot 0.7 \cdot 1499$
 $= 342628. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 273.05 \cdot (58.5 - 1.0) \cdot 0.74 \cdot 1499$



= 273563. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (31503 + 342628) = 374131 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (31503 + 0 + 273563 + 0) = 305066 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (31503 + 273563 + 0) = 305066 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 374131 kgf, must exceed W = 176970 kgf or W1 = 287947 kgf
 Path 2-2 = 305065 kgf, must exceed W = 176970 kgf or W2 = 425853 kgf
 Path 3-3 = 305065 kgf, must exceed W = 176970 kgf or W3 = 425853 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6094 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 3.6982 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 312.1982 mm

Input Echo, WRC107/537 Item 1, Description: D-10" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	4950.000	mm
Vessel Thickness	Tv	58.500	mm
Design Temperature	T1	430.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1498.98	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	406.647	mm
Nozzle Thickness	Tn	80.000	mm
Nozzle Material		SA-336 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1509.10	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1498.98	kgf/cm ²
Design Internal Pressure	Dp	5958.832	kPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

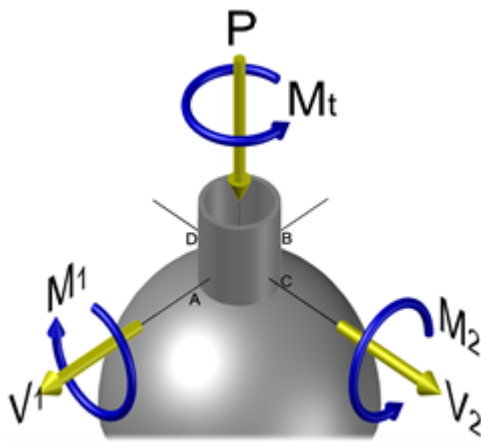
Radial Load (SUS)	P	2293.6	kgf
Longitudinal Shear (SUS)	(V1) V1	1656.5	kgf
Circumferential Shear (SUS)	(Vc) V2	1656.5	kgf
Circumferential Moment (SUS)	(Mc) M1	1274.3	kgf-m.

Longitudinal Moment	(SUS)	(M1)	M2	1274.3	kgf-m.
Torsional Moment	(SUS)		Mt	1465.4	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 406.647 + 2 \cdot 1.65 \cdot \sqrt{(2504.75 (58.5 - 1.0))}$$

$$= 1659.009 \text{ mm}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load		P	2293.6	kgf
Circumferential Shear	(VC)	V2	1656.5	kgf
Longitudinal Shear	(VL)	V1	1656.5	kgf
Circumferential Moment	(MC)	M1	1274.3	kgf-m.
Longitudinal Moment	(ML)	M2	1274.3	kgf-m.
Torsional Moment		MT	1465.4	kgf-m.

Dimensionless Param: U = 0.54 TAU = 5.00 (2.04) RHO = 0.72

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.06995	(A,B,C,D)
$M(x) / P$	SP 1	0.08420	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.10941	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.20830	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.10941	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.20830	(A,B,C,D)
$N(y) * T / P$	SP 1	0.12239	(A,B,C,D)
$M(y) / P$	SP 1	0.02840	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.09592	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.08236	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.09592	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.08236	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-4.9	-4.9	-4.9	-4.9	-4.9	-4.9	-4.9	-4.9
Rad. Bend. P		-35.0	35.0	-35.0	35.0	-35.0	35.0	-35.0	35.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-11.1	-11.1	11.1	11.1
Rad. Bend. MC		0.0	0.0	0.0	0.0	-126.9	126.9	126.9	-126.9
Rad. Memb. ML		-11.1	-11.1	11.1	11.1	0.0	0.0	0.0	0.0
Rad. Bend. ML		-126.9	126.9	126.9	-126.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-177.9	146.0	98.1	-85.6	-177.9	146.0	98.1	-85.6
Tang. Memb. P		-8.5	-8.5	-8.5	-8.5	-8.5	-8.5	-8.5	-8.5
Tang. Bend. P		-11.8	11.8	-11.8	11.8	-11.8	11.8	-11.8	11.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-9.7	-9.7	9.7	9.7
Tang. Bend. MC		0.0	0.0	0.0	0.0	-50.2	50.2	50.2	-50.2
Tang. Memb. ML		-9.7	-9.7	9.7	9.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-50.2	50.2	50.2	-50.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-80.2	43.8	39.6	-37.1	-80.2	43.8	39.6	-37.1
Shear VC		4.5	4.5	-4.5	-4.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.5	-4.5	4.5	4.5
Shear MT		9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8
Tot. Shear		14.3	14.3	5.3	5.3	5.3	5.3	14.3	14.3
Str. Int.		180.0	148.0	98.6	86.2	178.2	146.3	101.5	89.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1308.8	1308.8	1308.8	1308.8	1308.8	1308.8	1308.8	1308.8
Rad. Pl (SUS)		-16.0	-16.0	6.3	6.3	-16.0	-16.0	6.3	6.3
Rad. Q (SUS)		-162.0	162.0	91.9	-91.9	-162.0	162.0	91.9	-91.9
Long. Pm (SUS)		1308.8	1308.8	1308.8	1308.8	1308.8	1308.8	1308.8	1308.8
Long. Pl (SUS)		-18.2	-18.2	1.3	1.3	-18.2	-18.2	1.3	1.3
Long. Q (SUS)		-62.0	62.0	38.4	-38.4	-62.0	62.0	38.4	-38.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.5	4.5	-4.5	-4.5	-4.5	-4.5	4.5	4.5
Shear Q (SUS)		9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8
Pm (SUS)		1308.8	1308.8	1308.8	1308.8	1308.8	1308.8	1308.8	1308.8
Pm+Pl (SUS)		1296.3	1296.3	1317.7	1317.7	1296.3	1296.3	1317.7	1317.7
Pm+Pl+Q (Total)		1230.6	1456.7	1407.4	1272.2	1228.8	1455.0	1410.2	1275.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1308.75	1498.98	Passed
Pm+Pl (SUS)	1317.67	2248.47	Passed



Pm+Pl+Q (TOTAL)	1456.73	4496.94	Passed
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*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: MH-3-24"

From: 50

Pressure for Reinforcement Calculations	P	5941.943	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1049.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		220.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-3-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 1.181 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5941.94 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5941.94)$
 $= 57.0038$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5941.94 \cdot 304.8 / (1499 \cdot 1.0 + 0.4 \cdot 5941.94)$
 $= 12.1247$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.8178 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1099.2000	mm
Parallel to Vessel Wall, opening length	d	549.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		133.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	313.293	58.901	NA
Area in Shell	A1	10.971	206.461	NA
Area in Nozzle Wall	A2	47.548	74.964	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	16.674	16.674	NA
Area in Element	A5	255.199	255.199	NA
TOTAL AREA AVAILABLE	Atot	330.392	553.299	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness:	1020.1177	58.0000	mm
Based on given Pad Diameter:	1049.6000	54.1137	mm
Based on the Estimated Diameter Limit:	1097.6124	48.7898	mm

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (549.6•57.0038•1.0+2•30.0•57.0038•1.0•(1-1.))
= 313.293 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 549.6(1.0 • 59.0 - 1.0 • 57.004) - 2 • 30.0
(1.0 • 59.0 - 1.0 • 57.0038) • (1 - 1.)
= 10.971 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:
= (2 • Tlwp) • (tn - trn) • fr2
= (2 • 133.0) • (30.0 - 12.12) • 1.
= 47.548 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 1. + (0.0) • 1. + 1567.3604² • 1.
= 16.674 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1049.6 - 609.6) • 58.0 • 1.
= 255.199 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 12.1247 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 12.1247 mm

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, tg = 30.0, tr = 12.045, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 30.0, tr = 12.045, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 30.0, tr = 12.045, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Shell to Pad Weld Junction at Pad OD, Curve: A



Govrn. thk, tg = 58.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.96$, Temp. Reduction = 2 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A

Govrn. thk, tg = 30.0, tr = 12.045, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.401$, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Gov. MDMT of the Nozzle	:	-27 °C
Gov. MDMT of the Reinforcement Pad	:	37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5941.94/10340.00 = 0.575$

Weld Size Calculations, Description: MH-3-24"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (313.2928 - 10.9713 + 2 \cdot 30.0 \cdot 1. \cdot (1.0 \cdot 59.0 - 57.0038)) \cdot 1499)$
 $= 454968.69 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (47.5481 + 255.1995 + 16.6736 - 0.0 \cdot 1.) \cdot 1499$
 $= 478805.19 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (47.5481 + 0.0 + 1. + (35.3999)) \cdot 1499$
 $= 125836.26 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (47.5481 + 0.0 + 16.6736 + 255.1995 + (35.3999)) \cdot 1499$
 $= 531868.94 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1499$
 $= 70332. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.1416/2.0) \cdot 1049.6 \cdot 44.0 \cdot 0.49 \cdot 1499$
= 532828. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.1416 \cdot 289.8) \cdot (30.0 - 0.0) \cdot 0.7 \cdot 1499$
= 286591. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.1416/2) \cdot 609.6 \cdot 58.0 \cdot 0.74 \cdot 1499$
= 616054. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 609.6 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 1499$
= 616055. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (532828 + 286591) = 819418$ kgf

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(70332 + 616054 + 616055 + 0) = 1302442$ kgf

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(532828 + 616055 + 0) = 1148883$ kgf

Summary of Failure Path Calculations:

Path 1-1 = 819418 kgf, must exceed W = 454968 kgf or W1 = 478805 kgf
Path 2-2 = 1302441 kgf, must exceed W = 454968 kgf or W2 = 125836 kgf
Path 3-3 = 1148883 kgf, must exceed W = 454968 kgf or W3 = 531868 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6080 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 34.2085 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 594.2085 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 5.176 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: E-8"

From: 50

Pressure for Reinforcement Calculations	P	5949.491	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1009.30	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	819.0750	mm
Thickness of Pad	te	62.0000	mm
Weld leg size between Pad and Shell	Wp	60.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	60.0000	mm
Reinforcing Pad Width		300.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: E-8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 8.625 in.
Actual Thickness Used in Calculation 0.793 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5949.49 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5949.49)$
 $= 57.0780$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5949.49 \cdot 109.5375 / (1009 \cdot 1.0 + 0.4 \cdot 5949.49)$
 $= 6.4297$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.0023 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 357.6066 mm
Parallel to Vessel Wall, opening length d 178.8033 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 112.3396 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009.3 / 1499.0)$
 $= 0.673$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009.3 / 1499.0)$
 $= 0.673$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.673, 1.)$
 $= 0.673$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	109.566	20.572	NA
Area in Shell	A1	3.184	62.227	NA
Area in Nozzle Wall	A2	20.735	28.946	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.693	2.693	NA
Area in Element	A5	85.889	85.889	NA
TOTAL AREA AVAILABLE	Atot	112.501	179.755	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.



SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 352.8727 62.0000 mm
 Based on given Pad Diameter: 819.0750 59.8814 mm
 Based on the Estimated Diameter Limit: 356.0191 60.5755 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (178.8033 \cdot 57.078 \cdot 1.0 + 2 \cdot 20.1358 \cdot 57.078 \cdot 1.0 \cdot (1 - 0.67))$
 $= 109.566 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 178.803(1.0 \cdot 59.0 - 1.0 \cdot 57.078) - 2 \cdot 20.136$
 $(1.0 \cdot 59.0 - 1.0 \cdot 57.078) \cdot (1 - 0.673)$
 $= 3.184 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 112.34) \cdot (20.14 - 6.43) \cdot 0.6733$
 $= 20.735 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-can}/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 20.0^2 \cdot 0.67 + (0.0)^2 \cdot 0.67 + 0.0^2 \cdot 1.$
 $= 2.693 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (357.6066 - 219.075) \cdot 62.0 \cdot 1.$
 $= 85.889 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.4297 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 58.0780 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 58.0780 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9685 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 7.1600 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.16, \max(58.078, 2.5)]$
 $= 7.1600 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(6.4297, 7.16)$
 $= 7.1600 \text{ mm}$

Available Nozzle Neck Thickness = 20.1358 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 336.5,	Allowable	: 1009.3 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2429.1 kgf/cm ²	Passed
Occasional	: 121.1,	Allowable	: 1342.4 kgf/cm ²	Passed
Shear	: 125.9,	Allowable	: 706.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20.136, tr = 5.414, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 20.136, tr = 5.414, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 20.136, tr = 5.414, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 13 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 60.0, tr = 56.686, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.961$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 38 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 20.136, tr = 5.414, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : 38 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5949.49 / 10340.00 = 0.575$

Weld Size Calculations, Description: E-8"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$13.3000 = 0.7 \cdot t_{min}$	$14.1400 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$42.4200 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) S_v)$
 $= \max(0, (109.5664 - 3.1838 + 2 \cdot 20.1358 \cdot 0.6733 \cdot (1.0 \cdot 59.0 - 57.078)) 1499)$
 $= 160246.42 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (20.7349 + 85.8894 + 2.6933 - 0.0 \cdot 0.67) \cdot 1499$$

$$= 163864.67 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (20.7349 + 0.0 + 2.6933 + (15.9984)) \cdot 1499$$

$$= 59099.57 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (20.7349 + 0.0 + 2.6933 + 85.8894 + (15.9984)) \cdot 1499$$

$$= 187845.92 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 20.0 \cdot 0.49 \cdot 1009$$

$$= 34038. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 819.075 \cdot 60.0 \cdot 0.49 \cdot 1499$$

$$= 567003. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 99.4696) \cdot (20.1358 - 0.0) \cdot 0.7 \cdot 1009$$

$$= 44456. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 60.0 \cdot 0.74 \cdot 1499$$

$$= 229029. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (60.0 - 1.0) \cdot 0.74 \cdot 1499$$

$$= 225212. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (567003 + 44456) = 611459 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (34038 + 229029 + 225212 + 0) = 488278 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (567003 + 225212 + 0) = 792215 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 611458 kgf, must exceed W = 160246 kgf or W1 = 163864 kgf
 Path 2-2 = 488278 kgf, must exceed W = 160246 kgf or W2 = 59099 kgf
 Path 3-3 = 792214 kgf, must exceed W = 160246 kgf or W3 = 187845 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6009 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 4.3700 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 564.3700 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 178.80 mm

Input Echo, WRC297 Item 1, Description: E-8" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2750.000	mm	
Shell Thickness	60.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1498.98	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	219.075	mm	
Nozzle Thickness	20.1358	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-335 P11		
Nozzle Cold S.I. Allowable	SNmc	1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1009.30	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	819.075	mm
Thickness of Reinforcing Pad	62.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	5949.49	kPa
Radial Load	P	1834.88	kgf
Circumferential Shear	Vc	1325.19	kgf
Longitudinal Shear	Vl	1325.19	kgf
Circumferential Moment	Mc	815.53	kgf-m.
Longitudinal Moment	Ml	815.53	kgf-m.
Torsional Moment	Mt	937.86	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.651) is < 5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1436.5(60.0 - 1.0))}$$

$$= 1179.786 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2873.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	121.000	mm
Nozzle Thickness used	(t) =	20.136	mm
T / t	=	6.009	



$$d / t = 10.880$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.372$$

Nr/P	=	0.180
Mr/P	=	0.111
M0/P	=	0.269
N0/P	=	0.270
MrD/Mc	=	0.111
NrDL/Mc	=	0.153
M0d/Mc	=	0.300
N0DL/Mc	=	-0.117
MrD/Ml	=	0.110
NrDL/Ml	=	0.155
M0D/Ml	=	0.293
N0DL/Ml	=	0.083

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-20	20	-20	20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-44	44	44	-44
Normal Pressure Stress	660	721	660	721
Outplane Stress Summary	590	780	683	695

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-16	16	16	-16
Inplane Pressure Stress	330	330	330	330
Inplane Stress Summary	299	349	340	323

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10
Shear Stress Summary	13	13	7	7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	591	781	684	695

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-20	20	-20	20
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-45	45	45	-45
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	330	330	330	330

Outplane Stress Summary	263	395	349	298
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-16	16	16	-16
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	660	721	660	721

Inplane Stress Summary	629	740	671	714
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	10	10	10	10

Shear Stress Summary	7	7	13	13
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	629	740	671	714

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	290	363	290	363

Hoop Stress Summary	285	357	289	361
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-57	57	-57	57
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-142	-142	142	142

Axial Bending (ML)	-180	180	180	-180
Axial Pressure Stress	145	145	145	145
Axial Stress Summary	-249	226	395	149

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	19	19	-19	-19
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	61	61	61	61
Shear Stress Summary	80	80	42	42

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	558	396	410	370

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	290	363	290	363
Hoop Stress Summary	290	362	284	356

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	-14	-14	-14	-14
Axial Bending (P)	-57	57	-57	57
Axial Membrane (Mc)	-142	-142	142	142
Axial Bending (Mc)	-188	188	188	-188
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	145	145	145	145
Axial Stress Summary	-257	235	403	141

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-19	-19	19	19
Shear due to Torsion	61	61	61	61
Shear Stress Summary	42	42	80	80

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	554	375	444	383

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{1405.5(60.0 - 1.0)}$$

$$= 1169.363 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2811.000 mm
 Nozzle Outside Diameter (d) = 819.075 mm
 Vessel Thickness used (T) = 59.000 mm
 Nozzle Thickness used (t) = 320.136 mm
 T / t = 0.184
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.011$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.072
 Mr/P = 0.145
 M0/P = 0.043
 N0/P = 0.075
 MrD/Mc = 0.289
 NrDL/Mc = 0.093
 M0d/Mc = 0.088
 N0DL/Mc = 0.075
 MrD/Ml = 0.110
 NrDL/Ml = 0.091
 M0D/Ml = 0.034
 N0DL/Ml = 0.098

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-5	5	5	-5
Normal Pressure Stress	1385	1445	1385	1445
Outplane Stress Summary	1359	1458	1376	1452

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-45	45	-45	45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-18	18	18	-18
Inplane Pressure Stress	692	692	692	692
Inplane Stress Summary	621	751	664	718

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0

Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	3	3	0	0

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1359	1458	1376	1452

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	692	692	692	692
Outplane Stress Summary	658	715	692	689

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-45	45	-45	45
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-49	49	49	-49
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1385	1445	1385	1445
Inplane Stress Summary	1283	1534	1387	1441

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	3	3

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1283	1534	1387	1441

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	660	721	660	721	660	721	660	721



Circ. Pl (SUS)	-5	-5	-1	-1	-5	-5	1	1
Circ. Q (SUS)	-64	64	24	-24	-24	24	8	-8
Long. Pm (SUS)	330	330	330	330	330	330	330	330
Long. Pl (SUS)	-5	-5	1	1	-1	-1	-5	-5
Long. Q (SUS)	-24	24	8	-8	-65	65	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	660.0	721.0	660.0	721.0	660.0	721.0	660.0	721.0
Pm+Pl (SUS)	655.0	716.0	659.0	720.0	655.0	716.0	661.0	722.0
Pm+Pl+Q (Total)	591.6	780.4	683.1	696.1	631.1	740.1	669.5	714.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	721.00	1498.98	Passed
Pm+Pl (SUS)	722.02	2248.47	Passed
Pm+Pl+Q (TOTAL)	780.39	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1385	1445	1385	1445	1385	1445	1385	1445
Circ. Pl (SUS)	-5	-5	-1	-1	-5	-5	-1	-1
Circ. Q (SUS)	-18	18	-8	8	-94	94	4	-4
Long. Pm (SUS)	692	692	692	692	692	692	692	692
Long. Pl (SUS)	-5	-5	-1	-1	-5	-5	-1	-1
Long. Q (SUS)	-63	63	-27	27	-28	28	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	1385	1445	1385	1445	1385	1445	1385	1445
Pm+Pl (SUS)	1380	1440	1384	1444	1380	1440	1384	1444
Pm+Pl+Q (Total)	1362	1458	1376	1452	1286	1534	1388	1440

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1445.00	1498.98	Passed
Pm+Pl (SUS)	1444.00	2248.47	Passed
Pm+Pl+Q (TOTAL)	1534.00	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	290	363	290	363	290	363	290	363
Circ. Pl (SUS)	-5	-5	-1	-1	-1	-1	-5	-5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	145	145	145	145	145	145	145	145
Long. Pl (SUS)	-156	-156	128	128	-156	-156	128	128
Long. Q (SUS)	-237	237	123	-123	-245	245	131	-131
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19
Shear Q (SUS)	61	61	61	61	61	61	61	61
Pm (SUS)	290.0	363.0	290.0	363.0	290.0	363.0	290.0	363.0
Pm+Pl (SUS)	298.4	371.0	301.6	365.9	302.4	374.9	298.9	362.1
Pm+Pl+Q (Total)	556.5	395.7	410.5	370.0	551.4	374.6	444.2	384.4

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	363.00	1009.30	Passed
Pm+Pl (SUS)	374.93	1513.95	Passed
Pm+Pl+Q (TOTAL)	556.50	3027.90	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: UC-2-14"

From: 60

Pressure for Reinforcement Calculations	P	5929.490	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5250.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: UC-2-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 14.000 in.
Actual Thickness Used in Calculation 0.709 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5929.49 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5929.49 \cdot 177.8 / (1499 \cdot 1.0 + 0.4 \cdot 5929.49)$
 $= 7.0582$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	639.2000	mm
Parallel to Vessel Wall, opening length	d	319.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		103.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	181.793	34.252	NA
Area in Shell	A1	6.771	120.060	NA
Area in Nozzle Wall	A2	22.540	34.338	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	164.488	164.488	NA
TOTAL AREA AVAILABLE	Atot	194.799	319.886	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	616.7755	58.0000 mm
Based on given Pad Diameter:	655.5999	53.4139 mm
Based on the Estimated Diameter Limit:	637.6125	53.7146 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (319.6 \cdot 56.8813 \cdot 1.0 + 2 \cdot 18.0 \cdot 56.8813 \cdot 1.0 \cdot (1 - 1.))$
 $= 181.793 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 319.6(1.0 \cdot 59.0 - 1.0 \cdot 56.881) - 2 \cdot 18.0$
 $(1.0 \cdot 59.0 - 1.0 \cdot 56.8813) \cdot (1 - 1.)$
 $= 6.771 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 103.0) \cdot (18.0 - 7.06) \cdot 1.$
 $= 22.540 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (639.2 - 355.6) \cdot 58.0 \cdot 1.$
 $= 164.488 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 7.0582 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= ta$
 $= 7.0582 \text{ mm}$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, $tg = 18.0$, $tr = 7.012$, $c = 0.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, $tg = 18.0$, $tr = 7.012$, $c = 0.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $tg = 18.0$, $tr = 7.012$, $c = 0.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, $t_g = 58.0$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.957$, Temp. Reduction = 2°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, $t_g = 18.0$, $tr = 7.012$, $c = 0.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.39$, Temp. Reduction = 53°C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Gov. MDMT of the Nozzle	:	-44 °C
Gov. MDMT of the Reinforcement Pad	:	37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5929.49 / 10340.00 = 0.573$

Weld Size Calculations, Description: UC-2-14"

Intermediate Calc. for nozzle/shell Welds	T_{min}	18.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (181.7928 - 6.7712 + 2 \cdot 18.0 \cdot 1.0 \cdot (1.0 \cdot 59.0 - 56.8813)) \cdot 1499)$
 $= 263496.78 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (22.5401 + 164.4877 + 1.0 - 0.0 \cdot 1.0) \cdot 1499$
 $= 281849.47 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (22.5401 + 0.0 + 1.0 + (21.24)) \cdot 1499$
 $= 67124.34 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (22.5401 + 0.0 + 1.0 + 164.4877 + (21.24)) \cdot 1499$
 $= 313687.72 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 355.6 \cdot 10.0 \cdot 0.49 \cdot 1499$

= 41027. kgf

Shear, Pad Element Weld [Spew]:

= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.1416/2.0) \cdot 655.6 \cdot 44.0 \cdot 0.49 \cdot 1499$
= 332814. kgf

Shear, Nozzle Wall [Snw]:

= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.1416 \cdot 168.8) \cdot (18.0 - 0.0) \cdot 0.7 \cdot 1499$
= 100158. kgf

Tension, Pad Groove Weld [Tpgw]:

= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.1416/2) \cdot 355.6 \cdot 58.0 \cdot 0.74 \cdot 1499$
= 359365. kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 355.6 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 1499$
= 359366. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= $(SPEW + SNW) = (332814 + 100158) = 432973$ kgf

Failure Path [Path 2-2]:

= $(Sonw + Tpgw + Tngw + Sinw)$
= $(41027 + 359365 + 359366 + 0) = 759758$ kgf

Failure Path [Path 3-3]:

= $(Spew + Tngw + Sinw)$
= $(332814 + 359366 + 0) = 692180$ kgf

Summary of Failure Path Calculations:

Path 1-1 = 432972 kgf, must exceed W = 263496 kgf or W1 = 281849 kgf
Path 2-2 = 759757 kgf, must exceed W = 263496 kgf or W2 = 67124 kgf
Path 3-3 = 692179 kgf, must exceed W = 263496 kgf or W3 = 313687 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6129 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 11.5441 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5441 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot tnom / Rf \cdot (1 - Rf / Ro))$ 5.332 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: B-14"

From: 60

Pressure for Reinforcement Calculations	P	5929.490	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3650.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5929.49 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5929.49 \cdot 177.8 / (1499 \cdot 1.0 + 0.4 \cdot 5929.49)$
 $= 7.0582 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 639.2000 mm
 Parallel to Vessel Wall, opening length d 319.6000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 103.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	181.793	34.252	NA
Area in Shell	A1	6.771	120.060	NA
Area in Nozzle Wall	A2	22.540	34.338	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	164.488	164.488	NA
TOTAL AREA AVAILABLE	Atot	194.799	319.886	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.



SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 616.7755 58.0000 mm
 Based on given Pad Diameter: 655.5999 53.4139 mm
 Based on the Estimated Diameter Limit: 637.6125 53.7146 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (319.6 \cdot 56.8813 \cdot 1.0 + 2 \cdot 18.0 \cdot 56.8813 \cdot 1.0 \cdot (1 - 1.))$
 $= 181.793 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 319.6 (1.0 \cdot 59.0 - 1.0 \cdot 56.881) - 2 \cdot 18.0$
 $(1.0 \cdot 59.0 - 1.0 \cdot 56.8813) \cdot (1 - 1.)$
 $= 6.771 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 103.0) \cdot (18.0 - 7.06) \cdot 1.$
 $= 22.540 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (639.2 - 355.6) \cdot 58.0 \cdot 1.$
 $= 164.488 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 7.0582 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 57.8813 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 57.8813 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.9685 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.331, \max(57.8813, 2.5)]$
 $= 8.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(7.0582, 8.3312)$
 $= 8.3312 \text{ mm}$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 501.2,	Allowable	: 1499.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3258.9 kgf/cm ²	Passed
Occasional	: 254.1,	Allowable	: 1993.6 kgf/cm ²	Passed
Shear	: 135.7,	Allowable	: 1049.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)



Govrn. thk, tg = 18.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 18.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 18.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 58.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.957$, Temp. Reduction = 2 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 39 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 37 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 18.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Gov. MDMT of the Nozzle : -44 °C
 Gov. MDMT of the Reinforcement Pad : 37 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5929.49 / 10340.00 = 0.573$

Weld Size Calculations, Description: B-14"

Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	31.1080 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) Sv)$
 $= \max(0, (181.7928 - 6.7712 + 2 \cdot 18.0 \cdot 1. \cdot (1.0 \cdot 59.0 - 56.8813)) 1499)$
 $= 263496.78 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
= $(22.5401 + 164.4877 + 1. - 0.0 \cdot 1.) \cdot 1499$
= 281849.47 kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
= $(22.5401 + 0.0 + 1. + (21.24)) \cdot 1499$
= 67124.34 kgf

Weld Load [W3]:
= $(A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
= $(22.5401 + 0.0 + 1. + 164.4877 + (21.24)) \cdot 1499$
= 313687.72 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= $(\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
= $(3.1416/2.0) \cdot 355.6 \cdot 10.0 \cdot 0.49 \cdot 1499$
= 41027. kgf

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.1416/2.0) \cdot 655.6 \cdot 44.0 \cdot 0.49 \cdot 1499$
= 332814. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.1416 \cdot 168.8) \cdot (18.0 - 0.0) \cdot 0.7 \cdot 1499$
= 100158. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.1416/2) \cdot 355.6 \cdot 58.0 \cdot 0.74 \cdot 1499$
= 359365. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 355.6 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 1499$
= 359366. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (332814 + 100158) = 432973 \text{ kgf}$

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(41027 + 359365 + 359366 + 0) = 759758 \text{ kgf}$

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(332814 + 359366 + 0) = 692180 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 432972 kgf, must exceed W = 263496 kgf or W1 = 281849 kgf
Path 2-2 = 759757 kgf, must exceed W = 263496 kgf or W2 = 67124 kgf
Path 3-3 = 692179 kgf, must exceed W = 263496 kgf or W3 = 313687 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6129 kPa

Nozzle is O.K. for the External Pressure: 103 kPa



The Drop for this Nozzle is : 11.5441 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5441 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.332 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Input Echo, WRC297 Item 1, Description: B-14" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2750.000	mm
Shell Thickness	60.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1498.98	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	355.600	mm
Nozzle Thickness	18.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-387 11	
Nozzle Cold S.I. Allowable	SNmc 1509.10	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1498.98	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	655.600	mm
Thickness of Reinforcing Pad	58.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	5929.49	kPa
Radial Load	P	3211.03	kgf
Circumferential Shear	Vc	2319.08	kgf
Longitudinal Shear	Vl	2319.08	kgf
Circumferential Moment	Mc	2497.55	kgf-m.
Longitudinal Moment	Ml	2497.55	kgf-m.
Torsional Moment	Mt	2872.18	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{mean}(t - ca))}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(1434.5(60.0 - 1.0))}$
 $= 1315.642 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2869.000	mm
Nozzle Outside Diameter	(d) =	355.600	mm
Vessel Thickness used	(T) =	117.000	mm
Nozzle Thickness used	(t) =	18.000	mm
T / t	=	6.500	
d / t	=	19.756	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.614$$

Nr/P	=	0.093
Mr/P	=	0.060
M0/P	=	0.214
N0/P	=	0.349
MrD/Mc	=	0.095
NrDL/Mc	=	0.106
M0d/Mc	=	0.307
N0DL/Mc	=	0.124
MrD/Ml	=	0.094
NrDL/Ml	=	0.104
M0D/Ml	=	0.288
N0DL/Ml	=	0.216

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-30	30	-30	30
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-88	88	88	-88
Normal Pressure Stress	682	742	682	742

Outplane Stress Summary	544	842	743	686
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	340	340	340	340

Inplane Stress Summary	296	371	364	323
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	12	12	12	12

Shear Stress Summary	15	15	8	8
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	545	842	743	687
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-30	30	-30	30
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-94	94	94	-94
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	340	340	340	340

Outplane Stress Summary	201	451	403	274
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-29	29	29	-29
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	682	742	682	742

Inplane Stress Summary	636	772	706	725
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	12	12	12	12

Shear Stress Summary	8	8	15	15
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	637	772	706	725
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	-8	-8	-8	-8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	609	682	609	682

Hoop Stress Summary	590	663	612	685
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	-16	-16	-16	-16
Axial Bending (P)	-80	80	-80	80
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-162	-162	162	162
Axial Bending (ML)	-550	550	550	-550

Axial Pressure Stress	304	304	304	304
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Axial Stress Summary	-505	756	921	-19
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	23	23	-23	-23
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Shear due to (Vl)	0	0	0	0
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Shear due to Torsion	80	80	80	80
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Shear Stress Summary	103	103	57	57
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	1115	823	931	714
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-8	-8	-8	-8
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Hoop Bending (P)	0	0	0	0
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Hoop Membrane (Mc)	-6	-6	6	6
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Hoop Bending (Mc)	0	0	0	0
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Hoop Membrane (ML)	0	0	0	0
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Hoop Bending (ML)	0	0	0	0
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Hoop Pressure Stress	609	682	609	682
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Hoop Stress Summary	595	667	607	680
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	-16	-16	-16	-16
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Axial Bending (P)	-80	80	-80	80
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Axial Membrane (Mc)	-162	-162	162	162
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Axial Bending (Mc)	-543	543	543	-543
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Axial Membrane (ML)	0	0	0	0
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Axial Bending (ML)	0	0	0	0
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Axial Pressure Stress	304	304	304	304
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Axial Stress Summary	-498	748	913	-11
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
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Shear due to (Vl)	-23	-23	23	23
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Shear due to Torsion	80	80	80	80
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Shear Stress Summary	57	57	103	103
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	1099	778	945	722
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Stress Computations at the Edge of the Reinforcing Pad:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(1405.5(60.0 - 1.0))}$$

$$= 1305.888 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2811.000 mm
Nozzle Outside Diameter	(d) =	655.600 mm
Vessel Thickness used	(T) =	59.000 mm
Nozzle Thickness used	(t) =	168.000 mm
T / t	=	0.351
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.610$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.079
Mr/P	=	0.167
M0/P	=	0.049
N0/P	=	0.099
MrD/Mc	=	0.295
NrDL/Mc	=	0.092
M0d/Mc	=	0.091
N0DL/Mc	=	0.065
MrD/Ml	=	0.135
NrDL/Ml	=	0.084
M0D/Ml	=	0.043
N0DL/Ml	=	0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-9	-9	-9	-9
Outplane Bending (P)	-26	26	-26	26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	1380	1441	1380	1441
Outplane Stress Summary	1303	1473	1385	1443

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-7	-7	-7	-7
Inplane Bending (P)	-92	92	-92	92
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-88	88	88	-88
Inplane Pressure Stress	690	690	690	690
Inplane Stress Summary	492	854	688	696

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	7	7	7	7

Shear Stress Summary	10	10	3	3
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1303	1473	1385	1443
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-9	-9	-9	-9
Outplane Bending (P)	-26	26	-26	26
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	690	690	690	690

Outplane Stress Summary	587	761	721	654
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-7	-7	-7	-7
Inplane Bending (P)	-92	92	-92	92
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-193	193	193	-193
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1380	1441	1380	1441

Inplane Stress Summary	1077	1710	1484	1342
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	7	7	7	7

Shear Stress Summary	3	3	10	10
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1077	1710	1484	1342
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	B1	Cu	Cl	Du	D1
Circ. Pm (SUS)	682	742	682	742	682	742	682	742
Circ. Pl (SUS)	-19	-19	3	3	-7	-7	3	3



Circ. Q (SUS)	-118	118	58	-58	-37	37	21	-21
Long. Pm (SUS)	340	340	340	340	340	340	340	340
Long. Pl (SUS)	-7	-7	3	3	-14	-14	-2	-2
Long. Q (SUS)	-37	37	21	-21	-124	124	64	-64
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	682.0	742.0	682.0	742.0	682.0	742.0	682.0	742.0
Pm+Pl (SUS)	663.0	723.0	685.0	745.0	675.0	735.0	685.0	745.0
Pm+Pl+Q (Total)	545.9	841.5	743.2	687.2	638.2	772.3	706.7	724.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	742.00	1498.98	Passed
Pm+Pl (SUS)	745.02	2248.47	Passed
Pm+Pl+Q (TOTAL)	841.48	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1380	1441	1380	1441	1380	1441	1380	1441
Circ. Pl (SUS)	-22	-22	4	4	-16	-16	2	2
Circ. Q (SUS)	-53	53	1	-1	-285	285	101	-101
Long. Pm (SUS)	690	690	690	690	690	690	690	690
Long. Pl (SUS)	-16	-16	2	2	-16	-16	-2	-2
Long. Q (SUS)	-180	180	-4	4	-86	86	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	1380	1441	1380	1441	1380	1441	1380	1441
Pm+Pl (SUS)	1358	1419	1384	1445	1364	1425	1382	1443
Pm+Pl+Q (Total)	1305	1472	1385	1444	1079	1710	1483	1342

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1441.00	1498.98	Passed
Pm+Pl (SUS)	1445.01	2248.47	Passed
Pm+Pl+Q (TOTAL)	1710.02	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	609	682	609	682	609	682	609	682
Circ. Pl (SUS)	-19	-19	3	3	-14	-14	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	304	304	304	304	304	304	304	304
Long. Pl (SUS)	-178	-178	146	146	-178	-178	146	146
Long. Q (SUS)	-630	630	470	-470	-623	623	463	-463
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	23	23	-23	-23	-23	-23	23	23
Shear Q (SUS)	80	80	80	80	80	80	80	80
Pm (SUS)	609.0	682.0	609.0	682.0	609.0	682.0	609.0	682.0
Pm+Pl (SUS)	591.1	664.0	615.2	687.2	596.1	669.0	610.3	682.3
Pm+Pl+Q (Total)	1113	822	930	714	1097	778	944	722

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	682.00	1498.98	Passed
Pm+Pl (SUS)	687.23	2248.47	Passed
Pm+Pl+Q (TOTAL)	1113.23	4496.93	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: MH-2-24"

From: 80

Pressure for Reinforcement Calculations	P	5929.490	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		10550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1049.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		220.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-2-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 1.181 in.

Note:
Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5929.49 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5929.49 \cdot 304.8 / (1499 \cdot 1.0 + 0.4 \cdot 5929.49)$
 $= 12.0997$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.8178 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1099.2000	mm
Parallel to Vessel Wall, opening length	d	549.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		133.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	312.620	58.901	NA
Area in Shell	A1	11.644	206.461	NA
Area in Nozzle Wall	A2	47.615	74.964	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	16.674	16.674	NA
Area in Element	A5	255.199	255.199	NA
TOTAL AREA AVAILABLE	Atot	331.132	553.299	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness



Based on given Pad Thickness:	1017.6829	58.0000	mm
Based on given Pad Diameter:	1049.6000	53.7928	mm
Based on the Estimated Diameter Limit:	1097.6124	48.5004	mm

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1 - fr1)) UG-37(c)
= (549.6 • 56.8813 • 1.0 + 2 • 30.0 • 56.8813 • 1.0 • (1 - 1.))
= 312.620 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)
= 549.6 (1.0 • 59.0 - 1.0 • 56.881) - 2 • 30.0
(1.0 • 59.0 - 1.0 • 56.8813) • (1 - 1.)
= 11.644 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:
= (2 • Tlwp) • (tn - trn) • fr2
= (2 • 133.0) • (30.0 - 12.1) • 1.
= 47.615 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost) • Fr3 + ((Wi-can/0.707)² - Ar Lost) • fr2 + Trapfr4
= (1.) • 1. + (0.0) • 1. + 1567.3604² • 1.
= 16.674 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD)) • (min(tp,Tlwp,te)) • fr4
= (1049.6 - 609.6) • 58.0 • 1.
= 255.199 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 12.0997 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 12.0997 mm

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Shell to Pad Weld Junction at Pad OD, Curve: A



Govrn. thk, tg = 58.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.957$, Temp. Reduction = 2 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.401$, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Gov. MDMT of the Nozzle	:	-27 °C
Gov. MDMT of the Reinforcement Pad	:	37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5929.49/10340.00 = 0.573$

Weld Size Calculations, Description: MH-2-24"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (312.6199 - 11.6441 + 2 \cdot 30.0 \cdot 1. \cdot (1.0 \cdot 59.0 - 56.8813)) \cdot 1499)$
 $= 453061.59 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (47.6146 + 255.1995 + 16.6736 - 0.0 \cdot 1.) \cdot 1499$
 $= 478904.94 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (47.6146 + 0.0 + 1. + (35.3999)) \cdot 1499$
 $= 125935.98 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (47.6146 + 0.0 + 16.6736 + 255.1995 + (35.3999)) \cdot 1499$
 $= 531968.62 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1499$
 $= 70332. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.1416/2.0) \cdot 1049.6 \cdot 44.0 \cdot 0.49 \cdot 1499$
= 532828. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.1416 \cdot 289.8) \cdot (30.0 - 0.0) \cdot 0.7 \cdot 1499$
= 286591. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.1416/2) \cdot 609.6 \cdot 58.0 \cdot 0.74 \cdot 1499$
= 616054. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 609.6 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 1499$
= 616055. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (532828 + 286591) = 819418$ kgf

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(70332 + 616054 + 616055 + 0) = 1302442$ kgf

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(532828 + 616055 + 0) = 1148883$ kgf

Summary of Failure Path Calculations:

Path 1-1 = 819418 kgf, must exceed W = 453061 kgf or W1 = 478904 kgf
Path 2-2 = 1302441 kgf, must exceed W = 453061 kgf or W2 = 125935 kgf
Path 3-3 = 1148883 kgf, must exceed W = 453061 kgf or W3 = 531968 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6093 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 34.2085 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 594.2085 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 5.176 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: UC-1-14"

From: 90

Pressure for Reinforcement Calculations	P	5929.490	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12550.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: UC-1-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.709 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5929.49 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5929.49 \cdot 177.8 / (1499 \cdot 1.0 + 0.4 \cdot 5929.49)$
 $= 7.0582 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	639.2000	mm
Parallel to Vessel Wall, opening length	d	319.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		103.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	181.793	34.252	NA
Area in Shell	A1	6.771	120.060	NA
Area in Nozzle Wall	A2	22.540	34.338	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	164.488	164.488	NA
TOTAL AREA AVAILABLE	Atot	194.799	319.886	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	616.7755	58.0000 mm
Based on given Pad Diameter:	655.5999	53.4139 mm
Based on the Estimated Diameter Limit:	637.6125	53.7146 mm

Area Required [A]:
= ($d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)$) UG-37(c)
= ($319.6 \cdot 56.8813 \cdot 1.0 + 2 \cdot 18.0 \cdot 56.8813 \cdot 1.0 \cdot (1 - 1.)$)
= 181.793 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= $d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
= $319.6(1.0 \cdot 59.0 - 1.0 \cdot 56.881) - 2 \cdot 18.0$
= $(1.0 \cdot 59.0 - 1.0 \cdot 56.8813) \cdot (1 - 1.)$
= 6.771 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:
= $(2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
= $(2 \cdot 103.0) \cdot (18.0 - 7.06) \cdot 1.$
= 22.540 cm²

Area Available in Welds [A41 + A42 + A43]:
= $Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
= $10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
= 1.000 cm²

Area Available in Element [A5]:
= $(\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
= $(639.2 - 355.6) \cdot 58.0 \cdot 1.$
= 164.488 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 7.0582$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 7.0582 mm

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, $tg = 18.0$, $tr = 7.012$, $c = 0.0$ mm, $E* = 1.0$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, $tg = 18.0$, $tr = 7.012$, $c = 0.0$ mm, $E* = 1.0$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, $tg = 18.0$, $tr = 7.012$, $c = 0.0$ mm, $E* = 1.0$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.39$, Temp. Reduction = 53 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, $t_g = 58.0$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.957$, Temp. Reduction = 2°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, $t_g = 18.0$, $tr = 7.012$, $c = 0.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.39$, Temp. Reduction = 53°C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C

Gov. MDMT of the Nozzle	:	-44 °C
Gov. MDMT of the Reinforcement Pad	:	37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5929.49 / 10340.00 = 0.573$

Weld Size Calculations, Description: UC-1-14"

Intermediate Calc. for nozzle/shell Welds	T_{min}	18.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (181.7928 - 6.7712 + 2 \cdot 18.0 \cdot 1. \cdot (1.0 \cdot 59.0 - 56.8813)) \cdot 1499)$
 $= 263496.78 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (22.5401 + 164.4877 + 1. - 0.0 \cdot 1.) \cdot 1499$
 $= 281849.47 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (22.5401 + 0.0 + 1. + (21.24)) \cdot 1499$
 $= 67124.34 \text{ kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (22.5401 + 0.0 + 1. + 164.4877 + (21.24)) \cdot 1499$
 $= 313687.72 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 355.6 \cdot 10.0 \cdot 0.49 \cdot 1499$

= 41027. kgf

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 655.6 \cdot 44.0 \cdot 0.49 \cdot 1499 \\ &= 332814. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 168.8) \cdot (18.0 - 0.0) \cdot 0.7 \cdot 1499 \\ &= 100158. \text{ kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 355.6 \cdot 58.0 \cdot 0.74 \cdot 1499 \\ &= 359365. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 355.6 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 1499 \\ &= 359366. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (332814 + 100158) = 432973 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (41027 + 359365 + 359366 + 0) = 759758 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (332814 + 359366 + 0) = 692180 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 432972 kgf, must exceed W = 263496 kgf	or W1 = 281849 kgf
Path 2-2 = 759757 kgf, must exceed W = 263496 kgf	or W2 = 67124 kgf
Path 3-3 = 692179 kgf, must exceed W = 263496 kgf	or W3 = 313687 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6129 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 11.5441 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5441 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.332 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: A-10"

From: 100

Pressure for Reinforcement Calculations	P	5929.490	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		16900.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1009.30	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	593.0500	mm
Thickness of Pad	te	62.0000	mm
Weld leg size between Pad and Shell	Wp	46.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	60.0000	mm
Reinforcing Pad Width		160.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A-10"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 10.750 in.
 Actual Thickness Used in Calculation 0.984 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5929.49 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5929.49 \cdot 136.525 / (1009 \cdot 1.0 + 0.4 \cdot 5929.49)$
 $= 7.9876 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1401 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 446.0875 mm
 Parallel to Vessel Wall, opening length d 223.0437 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 124.5078 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009.3 / 1499.0)$
 $= 0.673$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1009.3 / 1499.0)$
 $= 0.673$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.673, 1.)$
 $= 0.673$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	136.162	25.655	NA
Area in Shell	A1	4.379	77.651	NA
Area in Nozzle Wall	A2	28.530	40.011	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.693	2.693	NA
Area in Element	A5	107.283	107.283	NA
TOTAL AREA AVAILABLE	Atot	142.885	227.639	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 435.2437 62.0000 mm
 Based on given Pad Diameter: 593.0500 58.1146 mm
 Based on the Estimated Diameter Limit: 444.5000 58.6527 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (223.0437 \cdot 56.8813 \cdot 1.0 + 2 \cdot 25.0031 \cdot 56.8813 \cdot 1.0 \cdot (1 - 0.67))$
 $= 136.162 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 223.044(1.0 \cdot 59.0 - 1.0 \cdot 56.881) - 2 \cdot 25.003$
 $(1.0 \cdot 59.0 - 1.0 \cdot 56.8813) \cdot (1 - 0.673)$
 $= 4.379 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 124.51) \cdot (25.0 - 7.99) \cdot 0.6733$
 $= 28.530 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-can}/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 20.0^2 \cdot 0.67 + (0.0)^2 \cdot 0.67 + 0.0^2 \cdot 1.$
 $= 2.693 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (446.0875 - 273.05) \cdot 62.0 \cdot 1.$
 $= 107.283 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 7.9876 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 57.8813 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 57.8813 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 1.9685 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.103, \max(57.8813, 2.5)]$
 $= 8.1026 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(7.9876, 8.1026)$
 $= 8.1026 \text{ mm}$

Available Nozzle Neck Thickness = 25.0031 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 295.6,	Allowable	: 1009.3 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2470.1 kgf/cm ²	Passed
Occasional	: 121.3,	Allowable	: 1342.4 kgf/cm ²	Passed
Shear	: 101.7,	Allowable	: 706.5 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B



Govrn. thk, tg = 25.003, tr = 6.726, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 25.003, tr = 6.726, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 25.003, tr = 6.726, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 19 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 60.0, tr = 56.491, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.957$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 38 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 25.003, tr = 6.726, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.269$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : 38 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5929.49 / 10340.00 = 0.573$

Weld Size Calculations, Description: A-10"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$13.3000 = 0.7 \cdot t_{min}$	$14.1400 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$32.5220 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) S_v)$
 $= \max(0, (136.1623 - 4.3794 + 2 \cdot 25.0031 \cdot 0.6733 \cdot (1.0 \cdot 59.0 - 56.8813)) 1499)$
 $= 198608.92 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (28.5297 + 107.283 + 2.6933 - 0.0 \cdot 0.67) \cdot 1499$$

$$= 207617.45 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (28.5297 + 0.0 + 2.6933 + (19.8656)) \cdot 1499$$

$$= 76580.61 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (28.5297 + 0.0 + 2.6933 + 107.283 + (19.8656)) \cdot 1499$$

$$= 237395.47 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 273.05 \cdot 20.0 \cdot 0.49 \cdot 1009$$

$$= 42424. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 593.05 \cdot 46.0 \cdot 0.49 \cdot 1499$$

$$= 314745. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 124.0234) \cdot (25.0031 - 0.0) \cdot 0.7 \cdot 1009$$

$$= 68828. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 273.05 \cdot 60.0 \cdot 0.74 \cdot 1499$$

$$= 285456. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 273.05 \cdot (60.0 - 1.0) \cdot 0.74 \cdot 1499$$

$$= 280699. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (314745 + 68828) = 383574 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (42424 + 285456 + 280699 + 0) = 608579 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (314745 + 280699 + 0) = 595445 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 383573 kgf, must exceed W = 198608 kgf or W1 = 207617 kgf
 Path 2-2 = 608578 kgf, must exceed W = 198608 kgf or W2 = 76580 kgf
 Path 3-3 = 595444 kgf, must exceed W = 198608 kgf or W3 = 237395 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6075 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 6.7946 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 566.7947 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 191.90 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 223.04 mm

Input Echo, WRC297 Item 1, Description: A-10" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2750.000	mm
Shell Thickness	60.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1498.98	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	25.0031	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-335 P11	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1009.30	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	593.050	mm
Thickness of Reinforcing Pad	62.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	5929.49	kPa
Radial Load	P	2293.59	kgf
Circumferential Shear	Vc	1656.48	kgf
Longitudinal Shear	Vl	1656.48	kgf
Circumferential Moment	Mc	1274.26	kgf-m.
Longitudinal Moment	Ml	1274.26	kgf-m.
Torsional Moment	Mt	1465.41	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.551) is < 5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1436.5(60.0 - 1.0))}$$

$$= 1233.761 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2873.000	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	121.000	mm
Nozzle Thickness used	(t) =	25.003	mm
T / t	=	4.839	



$$d / t = 10.921$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.463$$

Nr/P	=	0.180
Mr/P	=	0.116
M0/P	=	0.257
N0/P	=	0.263
MrD/Mc	=	0.117
NrDL/Mc	=	0.156
M0d/Mc	=	0.293
N0DL/Mc	=	-0.108
MrD/Ml	=	0.116
NrDL/Ml	=	0.158
M0D/Ml	=	0.285
N0DL/Ml	=	0.149

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-24	24	-24	24
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-54	54	54	-54
Normal Pressure Stress	658	719	658	719
Outplane Stress Summary	571	788	689	689

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-10	10	-10	10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	329	329	329	329
Inplane Stress Summary	288	354	342	320

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10
Shear Stress Summary	13	13	7	7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	571	789	689	689

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-24	24	-24	24
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-56	56	56	-56
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	329	329	329	329

Outplane Stress Summary	248	408	353	289
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-10	10	-10	10
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-22	22	22	-22
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	658	719	658	719

Inplane Stress Summary	617	744	672	709
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	10	10	10	10

Shear Stress Summary	7	7	13	13
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	617	744	672	710

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	291	363	291	363

Hoop Stress Summary	282	354	291	364
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-55	55	-55	55
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-114	-114	114	114

Axial Bending (ML)	-167	167	167	-167
Axial Pressure Stress	145	145	145	145
Axial Stress Summary	-204	242	360	136

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	15	15	-15	-15
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	49	49	49	49
Shear Stress Summary	65	65	34	34

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	504	384	374	369

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	291	363	291	363
Hoop Stress Summary	290	362	283	356

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-11	-11	-11	-11
Axial Bending (P)	-55	55	-55	55
Axial Membrane (Mc)	-114	-114	114	114
Axial Bending (Mc)	-176	176	176	-176
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	145	145	145	145
Axial Stress Summary	-213	251	369	127

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-15	-15	15	15
Shear due to Torsion	49	49	49	49
Shear Stress Summary	34	34	65	65

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	508	372	404	373

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1405.5(60.0 - 1.0))}$$

$$= 1223.338 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2811.000 mm
 Nozzle Outside Diameter (d) = 593.050 mm
 Vessel Thickness used (T) = 59.000 mm
 Nozzle Thickness used (t) = 185.003 mm
 T / t = 0.319
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.456$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.085
 Mr/P = 0.179
 M0/P = 0.052
 N0/P = 0.103
 MrD/Mc = 0.300
 NrDL/Mc = 0.088
 M0d/Mc = 0.092
 N0DL/Mc = 0.061
 MrD/M1 = 0.147
 NrDL/M1 = 0.089
 M0D/M1 = 0.046
 N0DL/M1 = 0.120

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-20	20	-20	20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	1380	1441	1380	1441
Outplane Stress Summary	1328	1464	1377	1445

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-70	70	-70	70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	690	690	690	690
Inplane Stress Summary	554	804	673	706

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	7	7	1	1

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1328	1464	1377	1445

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-20	20	-20	20
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	690	690	690	690
Outplane Stress Summary	624	734	700	673

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-70	70	-70	70
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-111	111	111	-111
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1380	1441	1380	1441
Inplane Stress Summary	1187	1611	1420	1400

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	1	1	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1187	1611	1420	1400

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	658	719	658	719	658	719	658	719

Circ. P1 (SUS)	-8	-8	0	0	-6	-6	2	2
Circ. Q (SUS)	-78	78	30	-30	-32	32	12	-12
Long. Pm (SUS)	329	329	329	329	329	329	329	329
Long. P1 (SUS)	-6	-6	2	2	-1	-1	-7	-7
Long. Q (SUS)	-32	32	12	-12	-80	80	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	658.0	719.0	658.0	719.0	658.0	719.0	658.0	719.0
Pm+P1 (SUS)	650.0	711.0	658.0	719.0	652.0	713.0	660.0	721.0
Pm+P1+Q (Total)	572.6	789.4	688.1	689.1	620.1	745.1	672.5	709.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	719.00	1498.98	Passed
Pm+P1 (SUS)	721.02	2248.47	Passed
Pm+P1+Q (TOTAL)	789.39	4496.94	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1380	1441	1380	1441	1380	1441	1380	1441
Circ. P1 (SUS)	-13	-13	1	1	-10	-10	0	0
Circ. Q (SUS)	-37	37	-3	3	-181	181	41	-41
Long. Pm (SUS)	690	690	690	690	690	690	690	690
Long. P1 (SUS)	-10	-10	0	0	-9	-9	-3	-3
Long. Q (SUS)	-124	124	-16	16	-54	54	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	1380	1441	1380	1441	1380	1441	1380	1441
Pm+P1 (SUS)	1367	1428	1381	1442	1370	1431	1380	1441
Pm+P1+Q (Total)	1330	1465	1378	1445	1189	1612	1421	1400

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1441.00	1498.98	Passed
Pm+P1 (SUS)	1442.01	2248.47	Passed
Pm+P1+Q (TOTAL)	1612.00	4496.94	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	291	363	291	363	291	363	291	363
Circ. Pl (SUS)	-8	-8	0	0	-1	-1	-7	-7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	145	145	145	145	145	145	145	145
Long. Pl (SUS)	-125	-125	103	103	-125	-125	103	103
Long. Q (SUS)	-222	222	112	-112	-231	231	121	-121
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	15	15	-15	-15	-15	-15	15	15
Shear Q (SUS)	49	49	49	49	49	49	49	49
Pm (SUS)	291.0	363.0	291.0	363.0	291.0	363.0	291.0	363.0
Pm+Pl (SUS)	283.9	355.7	295.7	364.9	290.8	362.7	289.4	358.0
Pm+Pl+Q (Total)	501.6	383.9	373.9	368.0	505.6	371.6	403.3	372.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	363.00	1009.30	Passed
Pm+Pl (SUS)	364.92	1513.95	Passed
Pm+Pl+Q (TOTAL)	505.59	3027.90	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-1-24"

From: 110

Pressure for Reinforcement Calculations	P	5929.490	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2750.00	mm
Design Length of Section	L	20278.3340	mm
Shell Finished (Minimum) Thickness	t	60.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		18910.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	1049.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		220.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 1.181 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5929.49 \cdot 1376.0002 / (1499 \cdot 1.0 - 0.6 \cdot 5929.49)$
 $= 56.8813$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5929.49 \cdot 304.8 / (1499 \cdot 1.0 + 0.4 \cdot 5929.49)$
 $= 12.0997$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.8178 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1099.2000	mm
Parallel to Vessel Wall, opening length	d	549.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		133.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1499.0 / 1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	312.620	58.901	NA
Area in Shell	A1	11.644	206.461	NA
Area in Nozzle Wall	A2	47.615	74.964	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	16.674	16.674	NA
Area in Element	A5	255.199	255.199	NA
TOTAL AREA AVAILABLE	Atot	331.132	553.299	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness



Based on given Pad Thickness:	1017.6829	58.0000	mm
Based on given Pad Diameter:	1049.6000	53.7928	mm
Based on the Estimated Diameter Limit:	1097.6124	48.5004	mm

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (549.6•56.8813•1.0+2•30.0•56.8813•1.0•(1-1.))
= 312.620 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 549.6(1.0 • 59.0 - 1.0 • 56.881) - 2 • 30.0
(1.0 • 59.0 - 1.0 • 56.8813) • (1 - 1.)
= 11.644 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:
= (2 • Tlwp) • (tn - trn) • fr2
= (2 • 133.0) • (30.0 - 12.1) • 1.
= 47.615 cm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (1.) • 1. + (0.0) • 1. + 1567.3604² • 1.
= 16.674 cm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1049.6 - 609.6) • 58.0 • 1.
= 255.199 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 12.0997 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 12.0997 mm

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Shell to Pad Weld Junction at Pad OD, Curve: A



Govrn. thk, tg = 58.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.957$, Temp. Reduction = 2 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A

Govrn. thk, tg = 30.0, tr = 12.02, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.401$, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Gov. MDMT of the Nozzle	:	-27 °C
Gov. MDMT of the Reinforcement Pad	:	37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5929.49/10340.00 = 0.573$

Weld Size Calculations, Description: MH-1-24"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (312.6199 - 11.6441 + 2 \cdot 30.0 \cdot 1. \cdot (1.0 \cdot 59.0 - 56.8813)) \cdot 1499)$
 $= 453061.59 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (47.6146 + 255.1995 + 16.6736 - 0.0 \cdot 1.) \cdot 1499$
 $= 478904.94 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (47.6146 + 0.0 + 1. + (35.3999)) \cdot 1499$
 $= 125935.98 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (47.6146 + 0.0 + 16.6736 + 255.1995 + (35.3999)) \cdot 1499$
 $= 531968.62 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 1499$
 $= 70332. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.1416/2.0) \cdot 1049.6 \cdot 44.0 \cdot 0.49 \cdot 1499$
= 532828. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.1416 \cdot 289.8) \cdot (30.0 - 0.0) \cdot 0.7 \cdot 1499$
= 286591. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.1416/2) \cdot 609.6 \cdot 58.0 \cdot 0.74 \cdot 1499$
= 616054. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 609.6 \cdot (58.0 - 0.0) \cdot 0.74 \cdot 1499$
= 616055. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (532828 + 286591) = 819418$ kgf

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(70332 + 616054 + 616055 + 0) = 1302442$ kgf

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(532828 + 616055 + 0) = 1148883$ kgf

Summary of Failure Path Calculations:

Path 1-1 = 819418 kgf, must exceed W = 453061 kgf or W1 = 478904 kgf
Path 2-2 = 1302441 kgf, must exceed W = 453061 kgf or W2 = 125935 kgf
Path 3-3 = 1148883 kgf, must exceed W = 453061 kgf or W3 = 531968 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6093 kPa

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 34.2085 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 594.2085 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 5.176 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

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Input, Nozzle Desc: C-14"

From: 120

Pressure for Reinforcement Calculations	P	5929.490	kPa
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	kPa
Temperature for External Pressure	Tempex	200	°C
Parent Material [Impact Tested]		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1498.98	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Elliptical Head	D	2750.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	58.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-387 11	
Material UNS Number		K11789	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	1498.98	kgf/cm ²
Allowable Stress At Ambient	Sna	1509.10	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Flange Material		SA-182 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	58.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-387 11	
Pad Allowable Stress at Temperature	Sp	1498.98	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1509.10	kgf/cm ²
Diameter of Pad along vessel surface	Dp	655.6000	mm
Thickness of Pad	te	58.0000	mm
Weld leg size between Pad and Shell	Wp	44.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	58.0000	mm
Reinforcing Pad Width		150.0000	mm
Flange Class		900	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: C-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.630 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 5929.49 \cdot 2752.0 \cdot 0.8994 / (2 \cdot 1498.98 \cdot 1.0 - 0.2 \cdot 5929.49)$
 $= 50.1217$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5929.49 \cdot 177.8 / (1499 \cdot 1.0 + 0.4 \cdot 5929.49)$
 $= 7.0582$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3308 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 647.2000 mm
 Parallel to Vessel Wall, opening length d 323.6000 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 98.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499.0/1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 1499.0/1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 1499.0/1499.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	162.194	12.202	NA
Area in Shell	A1	22.258	160.049	NA
Area in Nozzle Wall	A2	17.526	28.752	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	169.128	169.128	NA
TOTAL AREA AVAILABLE	Atot	209.912	358.928	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.



SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 564.9278 58.0000 mm
 Based on given Pad Diameter: 655.6000 41.6359 mm
 Based on the Estimated Diameter Limit: 645.6125 41.8638 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (323.6 \cdot 50.1217 \cdot 1.0 + 2 \cdot 16.0 \cdot 50.1217 \cdot 1.0 \cdot (1 - 1.))$
 $= 162.194 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 323.6(1.0 \cdot 57.0 - 1.0 \cdot 50.122) - 2 \cdot 16.0$
 $(1.0 \cdot 57.0 - 1.0 \cdot 50.1217) \cdot (1 - 1.)$
 $= 22.258 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 98.0) \cdot (16.0 - 7.06) \cdot 1.$
 $= 17.526 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-can}/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
 $= 1.000 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (647.2 - 355.6) \cdot 58.0 \cdot 1.$
 $= 169.128 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.0582 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 56.6755 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 56.6755 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.9672 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.331, \max(56.6755, 2.5)]$
 $= 8.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(7.0582, 8.3312)$
 $= 8.3312 \text{ mm}$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 564.8,	Allowable	: 1499.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3195.3 kgf/cm ²	Passed
Occasional	: 291.3,	Allowable	: 1993.6 kgf/cm ²	Passed
Shear	: 149.5,	Allowable	: 1049.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:A, Curve:B)



Govrn. thk, tg = 16.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.438$, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 16.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.438$, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: A

Govrn. thk, tg = 16.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.438$, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C

Shell to Pad Weld Junction at Pad OD, Curve: A

Govrn. thk, tg = 58.0, tr = 49.784, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.873$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 39 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 32 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 16.0, tr = 7.012, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.438$, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C

Gov. MDMT of the Nozzle : -36 °C
 Gov. MDMT of the Reinforcement Pad : 32 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5929.49 / 15510.00 = 0.382$

Weld Size Calculations, Description: C-14"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot \text{TminPad}$	$31.1080 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (162.1939 - 22.2581 + 2 \cdot 16.0 \cdot 1. \cdot (1.0 \cdot 57.0 - 50.1217)) 1499)$
 $= 213060.03 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
= $(17.5259 + 169.1276 + 1. - 0.0 \cdot 1.) \cdot 1499$
= 281288.56 kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
= $(17.5259 + 0.0 + 1. + (18.24)) \cdot 1499$
= 55111.28 kgf

Weld Load [W3]:
= $(A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
= $(17.5259 + 0.0 + 1. + 169.1276 + (18.24)) \cdot 1499$
= 308629.88 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= $(\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
= $(3.1416/2.0) \cdot 355.6 \cdot 10.0 \cdot 0.49 \cdot 1499$
= 41027. kgf

Shear, Pad Element Weld [Spew]:
= $(\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
= $(3.1416/2.0) \cdot 655.6 \cdot 44.0 \cdot 0.49 \cdot 1499$
= 332814. kgf

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.1416 \cdot 169.8) \cdot (16.0 - 0.0) \cdot 0.7 \cdot 1499$
= 89557. kgf

Tension, Pad Groove Weld [Tpgw]:
= $(\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
= $(3.1416/2) \cdot 355.6 \cdot 58.0 \cdot 0.74 \cdot 1499$
= 359365. kgf

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 355.6 \cdot (58.0 - 1.0) \cdot 0.74 \cdot 1499$
= 353170. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= $(SPEW + SNW) = (332814 + 89557) = 422371$ kgf

Failure Path [Path 2-2]:
= $(Sonw + Tpgw + Tngw + Sinw)$
= $(41027 + 359365 + 353170 + 0) = 753562$ kgf

Failure Path [Path 3-3]:
= $(Spew + Tngw + Sinw)$
= $(332814 + 353170 + 0) = 685984$ kgf

Summary of Failure Path Calculations:

Path 1-1 = 422371 kgf, must exceed W = 213060 kgf or W1 = 281288 kgf
Path 2-2 = 753561 kgf, must exceed W = 213060 kgf or W2 = 55111 kgf
Path 3-3 = 685983 kgf, must exceed W = 213060 kgf or W3 = 308629 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6070 kPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 kPa

The Drop for this Nozzle is : 6.2767 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 564.2767 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 4.711 %

Input Echo, WRC107/537 Item 1, Description: C-14 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	4950.000	mm
Vessel Thickness	Tv	58.000	mm
Design Temperature	T1	430.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1498.98	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

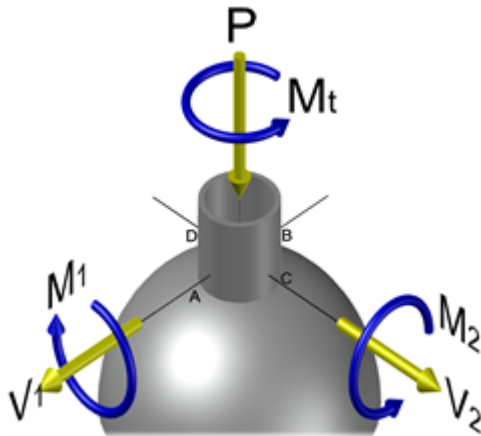
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	355.600	mm
Nozzle Thickness	Tn	16.000	mm
Nozzle Material		SA-387 11	
Nozzle UNS Number		K11789	
Nozzle Cold S.I. Allowable	SNmc	1509.10	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1498.98	kgf/cm ²
Thickness of Reinforcing Pad	Tpad	58.000	mm
Diameter of Reinforcing Pad	Dpad	655.600	mm
Design Internal Pressure	Dp	5929.490	kPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	3211.0	kgf
Longitudinal Shear (SUS)	(V1) V1	2319.1	kgf
Circumferential Shear (SUS)	(Vc) V2	2319.1	kgf
Circumferential Moment (SUS)	(Mc) M1	2497.5	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	2497.5	kgf-m.
Torsional Moment (SUS)	Mt	2872.2	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979
Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(2504.5 (58.0 - 1.0))}$$

$$= 1602.443 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	3211.0	kgf
Circumferential Shear	(VC) V2	2319.1	kgf
Longitudinal Shear	(VL) V1	2319.1	kgf
Circumferential Moment	(MC) M1	2497.5	kgf-m.
Longitudinal Moment	(ML) M2	2497.5	kgf-m.
Torsional Moment	MT	2872.2	kgf-m.

Dimensionless Param: U = 0.33 TAU = 10.61 RHO = 4.00 (7.19)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.06038	(A,B,C,D)
$M(x) / P$	SP 4	0.02476	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 4	0.18034	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 4	0.08006	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 4	0.18034	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 4	0.08006	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33991	(A,B,C,D)
$M(y) / P$	SP 4	0.12561	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 4	0.27645	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 4	0.46910	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 4	0.27645	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 4	0.46910	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
Rad.	Bend. P	-3.6	3.6	-3.6	3.6	-3.6	3.6	-3.6	3.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-6.3	-6.3	6.3	6.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-16.8	16.8	16.8	-16.8
Rad.	Memb. ML	-6.3	-6.3	6.3	6.3	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-16.8	16.8	16.8	-16.8	0.0	0.0	0.0	0.0

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7
Rad.	Bend. P	-24.8	24.8	-24.8	24.8	-24.8	24.8	-24.8	24.8
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-18.7	-18.7	18.7	18.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-117.0	117.0	117.0	-117.0
Rad.	Memb. ML	-18.7	-18.7	18.7	18.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-117.0	117.0	117.0	-117.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-168.2	115.4	103.2	-81.3	-168.2	115.4	103.2	-81.3
Tang. Memb. P		-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3
Tang. Bend. P		-7.7	7.7	-7.7	7.7	-7.7	7.7	-7.7	7.7
Tang. Memb. MC		0.0	0.0	0.0	0.0	-5.6	-5.6	5.6	5.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-35.0	35.0	35.0	-35.0
Tang. Memb. ML		-5.6	-5.6	5.6	5.6	0.0	0.0	0.0	0.0
Tang. Bend. ML		-35.0	35.0	35.0	-35.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-50.6	34.8	30.5	-24.1	-50.6	34.8	30.5	-24.1
Shear VC		4.0	4.0	-4.0	-4.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.0	-4.0	4.0	4.0
Shear MT		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5



Tot. Shear	11.4	11.4	3.5	3.5	3.5	3.5	11.4	11.4
Str. Int.	169.3	117.0	103.4	81.5	168.3	115.6	104.9	83.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		651.8	651.8	651.8	651.8	651.8	651.8	651.8	651.8
Rad. Pl (SUS)		-7.8	-7.8	4.8	4.8	-7.8	-7.8	4.8	4.8
Rad. Q (SUS)		-20.4	20.4	13.2	-13.2	-20.4	20.4	13.2	-13.2
Long. Pm (SUS)		651.8	651.8	651.8	651.8	651.8	651.8	651.8	651.8
Long. Pl (SUS)		-17.9	-17.9	1.4	1.4	-17.9	-17.9	1.4	1.4
Long. Q (SUS)		-116.8	116.8	80.2	-80.2	-116.8	116.8	80.2	-80.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		3.6	3.6	-3.6	-3.6	-3.6	-3.6	3.6	3.6
Shear Q (SUS)		12.6	12.6	12.6	12.6	12.6	12.6	12.6	12.6
Pm (SUS)		651.8	651.8	651.8	651.8	651.8	651.8	651.8	651.8
Pm+Pl (SUS)		645.2	645.2	659.0	659.0	645.2	645.2	659.0	659.0
Pm+Pl+Q (Total)		626.1	753.6	734.7	644.6	624.4	751.6	737.3	647.0

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	651.84	1498.98	Passed
Pm+Pl (SUS)	658.97	2248.47	Passed
Pm+Pl+Q (TOTAL)	753.62	4496.94	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		1313.7	1313.7	1313.7	1313.7	1313.7	1313.7	1313.7	1313.7
Rad. Pl (SUS)		-26.4	-26.4	11.0	11.0	-26.4	-26.4	11.0	11.0
Rad. Q (SUS)		-141.8	141.8	92.2	-92.2	-141.8	141.8	92.2	-92.2
Long. Pm (SUS)		1313.7	1313.7	1313.7	1313.7	1313.7	1313.7	1313.7	1313.7
Long. Pl (SUS)		-7.9	-7.9	3.2	3.2	-7.9	-7.9	3.2	3.2
Long. Q (SUS)		-42.7	42.7	27.3	-27.3	-42.7	42.7	27.3	-27.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.0	4.0	-4.0	-4.0	-4.0	-4.0	4.0	4.0
Shear Q (SUS)		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
Pm (SUS)		1313.7	1313.7	1313.7	1313.7	1313.7	1313.7	1313.7	1313.7
Pm+Pl (SUS)		1306.6	1306.6	1326.3	1326.3	1306.6	1306.6	1326.3	1326.3
Pm+Pl+Q (Total)		1264.2	1430.7	1417.1	1289.9	1263.2	1429.3	1418.7	1291.9



Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1313.73	1498.98	Passed
Pm+Pl (SUS)	1326.34	2248.47	Passed
Pm+Pl+Q (TOTAL)	1430.75	4496.94	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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